

JSS 1 SOLAR PHOTOVOLTAIC (PV) INSTALLATION AND MAINTENANCE

LESSON NOTES

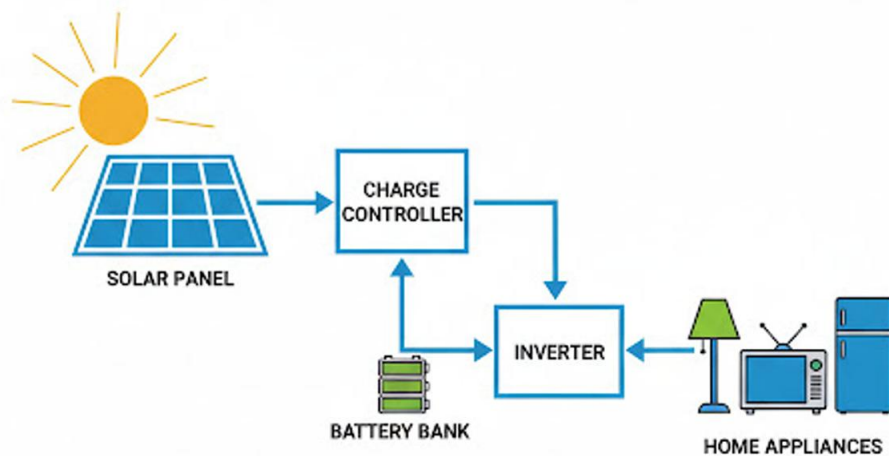
FIRST TERM

WEEK 1: INTRODUCTION TO SOLAR PHOTOVOLTAIC (PV) SYSTEM

1. Meaning and Importance of Solar Energy

Solar energy is electricity generated from sunlight using solar panels. It is a renewable and clean source of energy that never runs out as long as the sun shines.

SOLAR PV SYSTEM COMPONENTS



Importance:

- Reduces electricity bills significantly
- Provides reliable power supply during PHCN outages
- Environmentally friendly with no pollution
- Requires minimal maintenance once installed
- Creates job opportunities in installation and maintenance

2. Difference Between Solar and Other Sources of Energy

Feature	Solar Energy	Generator	PHCN Grid
Fuel Cost	None (free sunlight)	Requires petrol/diesel	Monthly bills
Noise	Silent operation	Very noisy	Silent
Maintenance	Low (cleaning only)	High (servicing, repairs)	None (by user)

Reliability	Works daily	Depends on fuel availability	Unstable supply
Environmental Impact	Clean, no pollution	Produces smoke/fumes	Some pollution
Lifespan	20-25 years	5-10 years	N/A

3. Components of a Basic PV System

- **Solar Panels:** Capture sunlight and convert it to DC electricity
- **Charge Controller:** Regulates power flow to batteries and prevents overcharging
- **Battery Bank:** Stores electricity for use at night or cloudy days
- **Inverter:** Converts DC electricity to AC electricity for home appliances
- **Cables and Connectors:** Link all components together safely
- **Mounting Structure:** Holds panels securely in position

4. Advantages of Solar Energy in Homes and Schools

For Homes:

- Eliminates or reduces monthly electricity bills
- Provides backup during power outages
- Increases property value
- Quiet and clean operation

For Schools:

- Reduces operating costs
- Enables uninterrupted learning
- Powers computers and projectors reliably
- Serves as a teaching tool for students
- Safe environment (no generator fumes)

Evaluation:

1. Define solar energy
2. State FOUR differences between solar energy and generator power
3. List and explain the function of FIVE components of a PV system
4. Mention THREE advantages of solar energy in schools

Assignment:

1. Draw and label a diagram of a basic solar PV system showing all components

2. Visit TWO locations in your area using solar panels and write a brief report on your observations



WEEK 2: ELECTRICAL APPLIANCES POWER LOAD / LOAD SIZING

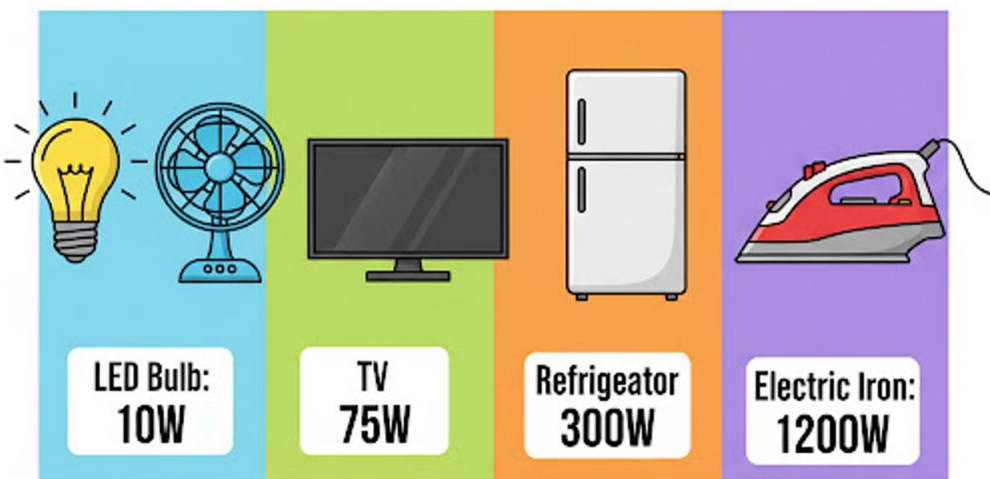
1. Meaning of Load

Load refers to electrical appliances or devices that consume electrical power. It is measured in Watts (W) or Kilowatts (kW).

- 1 Kilowatt (kW) = 1,000 Watts (W)

2. Identifying Electrical Appliances in Homes and Schools

APPLIANCE POWER RATINGS



Common Appliances and Their Power Ratings:

Appliance	Typical Power (W)
LED Bulb	5-15W
CFL Bulb	11-30W
Ceiling Fan	60-80W
Television (LED)	50-150W
Laptop	50-100W
Desktop Computer	200-400W
Refrigerator (Small)	100-150W
Air Conditioner (1HP)	1000-1200W
Electric Iron	1000-1500W
Phone Charger	5-10W
Decoder (DSTV)	15-30W
Microwave Oven	800-1500W
Water Pump (0.5HP)	400-600W

3. Calculating Total Power Load (Wattage)

Formula: Total Load = Sum of all appliances running at the same time

Example 1:

- 4 LED bulbs (10W each) = 40W
- 2 Fans (75W each) = 150W
- 1 TV (100W) = 100W
- 1 Decoder (20W) = 20W

Total Load = 40 + 150 + 100 + 20 = 310W

Example 2: A classroom with:

- 6 lights (15W each) = 90W
- 3 fans (70W each) = 210W
- 1 projector (300W) = 300W

Total Load = 90 + 210 + 300 = 600W

Important: Always add 20-25% extra capacity for safety and future expansion.

4. Estimating Energy Consumption Using Power Ratings

Formula: Energy (Wh) = Power (W) × Time (hours)

Example 1: A 60W fan runs for 8 hours daily:

- Daily Energy = 60W × 8h = 480Wh = 0.48kWh

Example 2: Calculate daily energy for a home:

Appliance	Power (W)	Hours/Day	Energy (Wh)
3 Bulbs	30W	6 hours	180Wh
2 Fans	140W	8 hours	1,120Wh
TV	100W	4 hours	400Wh
Refrigerator	150W	12 hours*	1,800Wh
TOTAL			3,500Wh = 3.5kWh

*Refrigerators cycle on/off, so they don't run continuously

Evaluation:

1. What is electrical load?
2. List EIGHT electrical appliances with their power ratings
3. Calculate total load: 5 bulbs (10W each), 3 fans (75W each), 1 TV (120W), 1 laptop (65W)
4. A 100W bulb runs 6 hours daily. Calculate weekly energy consumption

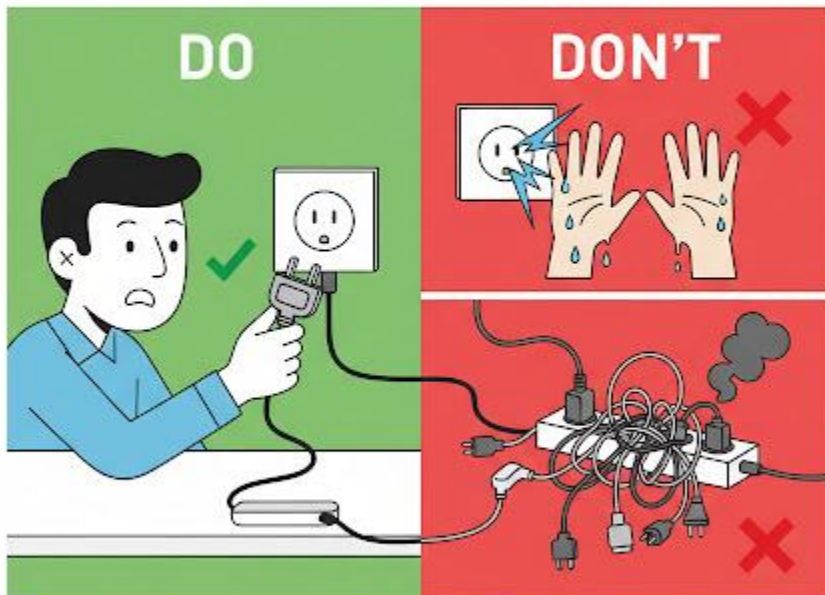
Assignment:

1. List all electrical appliances in your home with their power ratings (check labels or ask parents)
2. Calculate the total load if all appliances run simultaneously
3. Calculate daily energy consumption of your family's refrigerator



WEEK 3: ELECTRICAL SAFETY PRECAUTIONS

ELECTRICAL SAFETY DO vs. DON'T



1. Definition of Electrical Safety

Electrical safety refers to practices and precautions taken to prevent electrical accidents, injuries, or death when using or working with electricity.

2. Causes of Electrical Accidents

- Faulty wiring and loose connections
- Overloading electrical sockets
- Using damaged cables and plugs
- Touching live wires
- Wet hands or surfaces when handling electricity
- Poor installation by unqualified persons
- Lack of proper grounding/earthing

- Inserting metal objects into sockets
- Working on live circuits without training

3. Basic Rules for Safe Use of Electricity

1. **Never touch electrical devices with wet hands**
2. **Don't overload sockets** - use one high-power appliance per socket
3. **Switch off before unplugging** - turn off appliance, then remove plug
4. **Inspect cables regularly** - replace damaged or frayed cables immediately
5. **Keep electrical items away from water**
6. **Don't insert metal objects into sockets**
7. **Use proper fuses** - never use oversized fuses or wire substitutes
8. **Ensure proper earthing** - use three-pin plugs for metal appliances
9. **Turn off main power during storms** - protect against lightning surges
10. **Report electrical problems immediately** - tell adults about exposed wires or sparking outlets
11. **Don't attempt repairs without training** - hire qualified electricians
12. **Keep safe distance from power lines** - stay at least 3 meters away

4. First Aid Procedures for Electric Shock Victims

STEP 1: Ensure Your Safety

- DO NOT touch the victim if still in contact with electricity
- You will also be electrocuted

STEP 2: Cut Off Power Source

- Turn off the main switch or circuit breaker immediately
- If you can't reach the switch, use a dry wooden stick or plastic chair to push the victim away from the electrical source
- Never use metal objects or anything wet

STEP 3: Call for Help

- Shout for help immediately
- Call emergency services or send someone to get medical help

STEP 4: Check the Victim's Condition

- Check if they are breathing
- Check for pulse (heartbeat)

STEP 5: Provide First Aid

- **If breathing:** Place them on their side (recovery position)
- **If not breathing:** Begin CPR if trained (30 chest compressions, 2 rescue breaths)
- **If conscious:** Keep them calm, lying down, and warm
- Cover any burns with clean, dry cloth
- Do NOT apply ice, butter, or ointments to burns

STEP 6: Stay with Victim

- Monitor breathing until medical help arrives
- Keep them warm
- Reassure them

Important DON'Ts:

- Don't touch victim while connected to electricity
- Don't give them food or water
- Don't move them unnecessarily
- Don't apply ointments to burns

Evaluation:

1. Define electrical safety
2. List SIX causes of electrical accidents
3. State EIGHT rules for safe use of electricity
4. Describe the correct procedure for rescuing an electric shock victim
5. Why should you never touch an electrocuted person while they are still in contact with the electrical source?

Assignment:

1. Create a safety poster showing FIVE electrical safety rules
2. Inspect your home and list FIVE electrical hazards you find. Suggest how to fix each one
3. Interview an adult about an electrical accident they witnessed and write a report on what happened and how it could have been prevented

WEEK 4: SAFETY PRECAUTIONS IN HANDLING SOLAR PV SYSTEM MATERIALS

Battery Safety



1. Safe Handling of Solar Panels, Batteries, and Cables

A) SOLAR PANELS

Safety Precautions:

- Always carry panels with two people - they are heavy (15-25kg)
- Hold by the frame, not the cables
- Wear gloves to protect from sharp edges
- Panels produce electricity whenever exposed to light - cover them when working on connections
- Don't drop or hit panels - the glass can break
- Store panels upright or at an angle, never flat
- Use proper lifting technique - bend knees, not back

- Wear safety shoes when moving panels

B) BATTERIES

Batteries are the most dangerous component in solar systems.

Hazards:

- Contain sulfuric acid (highly corrosive)
- Produce explosive hydrogen gas when charging
- Can deliver very high currents causing severe burns
- Very heavy (25-70kg)

Safety Precautions:

- **Wear protective equipment:** goggles, acid-resistant gloves, apron, boots
- Install in well-ventilated areas (hydrogen gas is explosive)
- Never smoke or create sparks near batteries
- Connect correctly - positive to positive, negative to negative
- Remove metal jewelry before working on batteries
- Use insulated tools
- Keep batteries upright - tilting causes acid leakage
- If acid touches skin: wash immediately with plenty of water for 15 minutes
- If acid enters eyes: flush with water for 15 minutes, seek medical help immediately

C) CABLES

Safety Precautions:

- Use correct cable size for the current
- Protect cables in conduits where they can be damaged
- Avoid sharp bends
- Keep cables away from heat sources
- Use waterproof connectors for outdoor installations
- Secure cables properly - don't let them hang loose
- Check connections are tight
- Replace damaged cables immediately

2. Storage and Maintenance of Solar PV Tools

Common Tools:

- Multimeter (measures voltage, current)
- Wire strippers
- Screwdrivers (insulated)
- Wrenches and spanners
- Cable cutters
- Drill and bits
- Measuring tape

Storage:

- Keep tools in a dry toolbox
- Clean tools after each use
- Store sharp tools with covers
- Organize tools so each has a place
- Keep away from children
- Check tools before use

Maintenance:

- Inspect for damage before use
- Clean and oil metal parts to prevent rust
- Test electrical tools for proper function
- Sharpen cutting tools when dull
- Replace damaged tools
- Store batteries separately from power tools

3. Use of Safety Kits (Gloves, Goggles, Boots)

A) SAFETY GLOVES

Types:

- **Work gloves:** For handling panels and hardware - protect from cuts
- **Insulated rubber gloves:** For electrical work - protect from shock
- **Chemical-resistant gloves:** For battery work - protect from acid

Care: Inspect before use, replace if torn, store in clean place

B) SAFETY GOGGLES/FACE SHIELDS

When to wear:

- Always when working with batteries (acid splash protection)
- When drilling or grinding (flying particles)
- When working overhead (falling debris)

Care: Clean regularly, replace if scratched, store in protective case

C) SAFETY BOOTS

Features:

- Steel toe cap (protects from falling objects)
- Slip-resistant sole
- Ankle support
- Provides some electrical insulation

When to wear: Always during installation and maintenance work

D) OTHER SAFETY EQUIPMENT

- **Hard hat:** Protects head from falling objects
- **Safety harness:** For working on roofs above 2 meters
- **High-visibility vest:** Makes you visible on busy sites
- **Dust mask:** Protects lungs when drilling

4. Environmental Considerations in Handling PV Materials

Why It Matters: Solar energy is clean, but we must handle materials responsibly to protect the environment.

Environmental Practices:

Solar Panels:

- Handle carefully to maximize 25-year lifespan

- Recycle broken or old panels at e-waste facilities
- Don't throw in regular trash

Batteries:

- NEVER throw batteries in regular trash
- Lead and acid contaminate soil and water, causing serious health problems
- Always return old batteries to dealers or recycling centers
- Battery recycling recovers lead, acid, and plastic for reuse

Cables and Components:

- Recycle copper from old cables
- Take electronic waste (inverters, controllers) to proper recycling facilities
- Don't burn cables to recover copper (releases toxic fumes)

General:

- Reuse packaging materials when possible
- Use minimal water for panel cleaning
- Use environmentally friendly cleaning solutions
- Dispose of all materials properly

Benefits of Solar for Environment:

- Reduces carbon emissions
- No air or water pollution during operation
- Reduces dependence on fossil fuels
- Each home solar system offsets tons of CO₂ over its lifetime

Evaluation:

1. List FIVE safety precautions when handling solar panels
2. Why are batteries considered dangerous? Mention THREE hazards
3. What protective equipment should you wear when working with batteries?
4. Describe proper first aid if battery acid touches your skin
5. Name THREE types of safety gloves and when each should be used

6. Why is it important to recycle old batteries?
7. List FOUR tools used in solar installation and explain how to store each

Assignment:

1. Create a safety checklist for installing solar panels with at least 15 items organized by: before work, during installation, battery handling, and after completion
2. Design a chart showing different solar PV tasks and the required PPE for each (at least 5 tasks)
3. Write a one-page report on battery recycling: what happens to recycled batteries, where to recycle them in your area, and environmental problems prevented by recycling



WEEK 6: SOLAR PV COMPONENTS - SOLAR PANELS

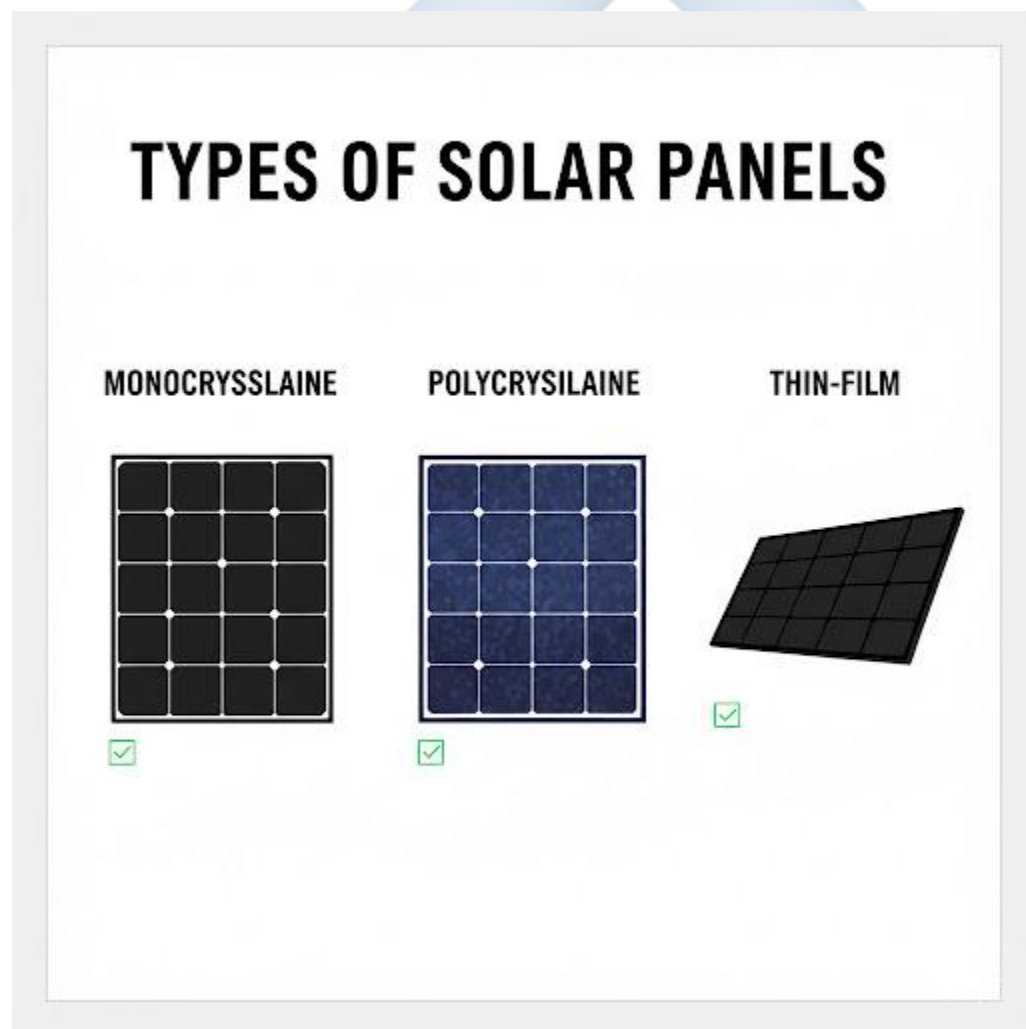
1. Meaning and Function of Solar Panels

Definition: Solar panels (photovoltaic modules or PV panels) are devices that convert sunlight directly into electricity through the photovoltaic effect.

How They Work:

- Solar panels contain many small units called solar cells
- When sunlight hits the solar cells, it excites electrons
- This movement of electrons creates direct current (DC) electricity
- Multiple cells are connected together to form a panel

2. Types of Solar Panels



A) MONOCRYSTALLINE SOLAR PANELS

Appearance:

- Black or dark blue color
- Rounded edges on cells
- Uniform appearance

Characteristics:

- Made from single pure silicon crystal
- Most efficient (15-22% efficiency)
- Perform better in low light and high temperatures
- Most expensive
- Space-efficient (need fewer panels)
- Lifespan: 25-30 years

Best For: Limited roof space, maximum power output

B) POLYCRYSTALLINE SOLAR PANELS

Appearance:

- Bright blue color
- Square cells with straight edges
- Speckled, fragmented appearance

Characteristics:

- Made from multiple silicon fragments melted together
- Moderate efficiency (13-16%)
- More affordable than monocrystalline
- Require more space for same power output
- Lifespan: 25-28 years

Best For: Budget-conscious installations with adequate space

C) THIN FILM SOLAR PANELS

Appearance:

- Uniform black or blue
- Sleek, smooth surface
- Flexible options available

Characteristics:

- Made by depositing thin layers of photovoltaic material
- Lowest efficiency (7-13%)
- Lightweight and flexible
- Least expensive
- Perform better in shade and high temperatures
- Require most space
- Shorter lifespan: 15-20 years

Best For: Large commercial installations, curved surfaces, portable applications

Comparison Table

Feature	Monocrystalline	Polycrystalline	Thin Film
Efficiency	15-22%	13-16%	7-13%
Cost	Highest	Moderate	Lowest
Space Needed	Least	Moderate	Most
Appearance	Black/Dark	Bright Blue	Black/Blue
Lifespan	25-30 years	25-28 years	15-20 years

3. Installation Position and Direction**A) DIRECTION (ORIENTATION)**

- **In Nigeria (Northern Hemisphere):** Panels should face **TRUE SOUTH** for maximum sun exposure
- Use a compass to find true south
- East or west-facing panels produce about 20% less power

- North-facing panels produce very little power

B) TILT ANGLE

Best tilt angle is approximately equal to your latitude:

- **Lagos (6°N):** 5-10 degree tilt
- **Abuja (9°N):** 10-15 degree tilt
- **Kano (12°N):** 12-17 degree tilt
- Minimum tilt of 10 degrees helps rainwater wash off dust

C) SHADING CONSIDERATIONS

- Panels must be free from shade throughout the day
- Even small shadows dramatically reduce output
- Check for shade from trees, buildings, chimneys, power lines
- Remember: sun position changes with seasons

D) MOUNTING LOCATION

- **Rooftop:** Most common, doesn't use ground space
- **Ground-mounted:** Easier maintenance, better cooling
- **Pole-mounted:** Good for small systems in open areas

E) STRUCTURAL CONSIDERATIONS

- Roof must be strong enough to support panels (15-25kg each plus mounting hardware)
- Mounting frame must be securely attached
- Allow air circulation underneath panels for cooling
- Ensure proper waterproofing where mounts penetrate roof

4. Cleaning and Maintenance of Panels

Why Cleaning is Important

- Dust, dirt, bird droppings, and leaves reduce power output by 15-25%
- Clean panels can produce up to 30% more power than dirty panels
- Nigeria's dusty harmattan season requires more frequent cleaning

Cleaning Schedule

- **Dry season/Harmattan:** Every 2-3 weeks
- **Rainy season:** Monthly (rain helps clean panels)
- **After dust storms:** As soon as possible
- **Bird droppings:** Clean immediately

Cleaning Procedure

Materials Needed:

- Soft brush or squeegee
- Mild soap or panel cleaning solution
- Clean water (preferably soft water)
- Hose or bucket
- Soft cloth or sponge

Steps:

1. Clean early morning or late evening (never clean hot panels)
2. Turn off the system if possible
3. Rinse with water first to remove loose dust
4. Apply mild soap solution using soft brush or cloth
5. Scrub gently in circular motions
6. Rinse thoroughly with clean water
7. Check for damage (cracks, broken glass, loose connections)
8. Dry or let air dry naturally

Safety During Cleaning:

- Use proper fall protection if on roof
- Wear non-slip shoes
- Don't walk on panels
- Be careful of electrical connections
- Work with a partner for rooftop cleaning

Things to AVOID

- ✗ Abrasive materials (steel wool, rough brushes) - scratch glass
- ✗ Harsh chemicals (bleach, strong detergents) - damage panel coating
- ✗ High-pressure washers - can damage seals and cells
- ✗ Cleaning during midday heat - thermal shock can crack glass
- ✗ Walking directly on panels - can cause micro-cracks

Other Maintenance Tasks

Monthly Inspections:

- Visual check for physical damage
- Check mounting structure for looseness
- Inspect cables for damage
- Look for vegetation growth blocking panels

Quarterly Checks:

- Test system output with multimeter
- Tighten loose bolts on mounting structure
- Check and clean connections
- Verify charge controller and inverter displays

Annual Maintenance:

- Professional inspection recommended
- Thermal imaging to detect problem cells
- Electrical testing of all components
- Complete system performance evaluation

Monitoring Performance:

- Keep records of daily power production
- Sudden drops indicate problems (shading, dirt, or component failure)
- Many modern systems have apps for monitoring

EVALUATION

1. What is a solar panel and what is its main function?
2. Compare monocrystalline and polycrystalline solar panels. State THREE differences.
3. Which type of solar panel is most efficient? Which is cheapest?
4. In which direction should solar panels face in Nigeria for maximum power?
5. Why is shading harmful to solar panel output?
6. Explain the correct procedure for cleaning solar panels.
7. How often should panels be cleaned during harmattan season?
8. List FOUR things to avoid when cleaning solar panels.

ASSIGNMENT

1. Draw and label the three types of solar panels, showing their distinctive appearances.
2. Use a compass to determine which direction is south. Observe your school or home roof and write a report on whether it would be suitable for solar panel installation (consider direction, shading, and structural strength).
3. Create a maintenance schedule for solar panels covering daily, weekly, monthly, and annual tasks.

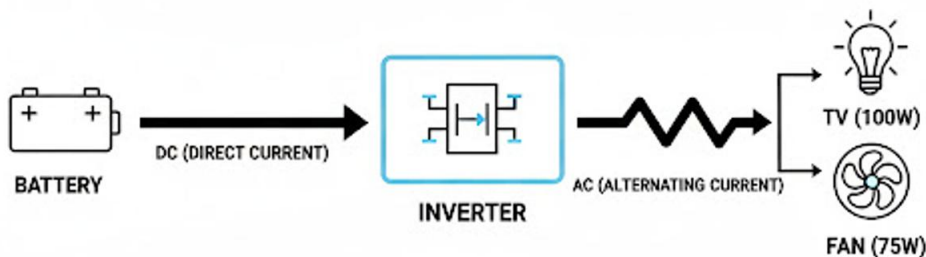
WEEK 8: SOLAR PV COMPONENTS - INVERTERS

1. Meaning and Function of an Inverter

Definition: An inverter is an electronic device that converts Direct Current (DC) electricity into Alternating Current (AC) electricity.

SOLAR PV COMPONENTS - INVERTERS

HOW AN INVERTER WORKS



Understanding Current Types:

- **DC (Direct Current):** Flows in one direction only. Produced by solar panels and batteries
- **AC (Alternating Current):** Changes direction rapidly (50 times per second in Nigeria). Used by most home appliances

Why We Need Inverters:

- Solar panels and batteries produce DC electricity
- Most appliances (TVs, fans, refrigerators) need AC electricity
- The inverter bridges this gap by converting DC to AC

Functions of an Inverter:

- Converts DC to AC electricity
- Regulates voltage (maintains steady 220-240V output)
- Controls frequency (maintains 50Hz in Nigeria)
- Protects against overload, short circuits, and over-temperature

- Charges batteries (in hybrid inverters)
- Displays system information

2. Types of Inverters

A) CLASSIFICATION BY WAVEFORM

i) Pure Sine Wave Inverters

- Produce smooth AC output identical to PHCN grid power
- Work with ALL appliances including sensitive electronics
- Most efficient for motors (fans, refrigerators, pumps)
- Most expensive
- **Best For:** Complete home systems, sensitive equipment

ii) Modified Sine Wave Inverters

- Produce stepped, approximated AC output
- Less expensive
- May cause humming in fans and fluorescent lights
- Some appliances may not work properly
- Less efficient for motors
- **Best For:** Basic lighting and simple appliances

iii) Square Wave Inverters

- Produce crude, blocky AC output
- Cheapest but lowest quality
- Many appliances won't work properly
- Rarely used today

B) CLASSIFICATION BY APPLICATION

i) Stand-Alone (Off-Grid) Inverters

- For systems not connected to PHCN grid
- Work independently with solar panels and batteries

- Most common in Nigeria
- **Uses:** Homes without grid access, rural areas

ii) Grid-Tie Inverters

- Connect directly to PHCN grid
- No batteries required
- Feed excess power to grid
- Stop working during power outages (safety feature)
- **Advantages:** Lower cost, can sell power to grid
- **Disadvantages:** No power during outages

iii) Hybrid Inverters

- Combine off-grid and grid-tie features
- Can work with batteries AND grid
- Provide backup during outages
- Most flexible but more expensive
- **Best For:** Areas with unstable grid supply

C) CLASSIFICATION BY SIZE

Small Inverters (100W - 500W):

- Power few devices (phones, laptops, small TV)
- Portable
- Good for travel or small applications

Medium Inverters (600W - 2kW):

- Power typical home lighting and appliances
- Most common for small homes

Large Inverters (2.5kW - 10kW+):

- Power entire homes including air conditioners
- Used in commercial installations

3. Conversion of DC to AC Power

How Inverters Work (Simplified):

1. **DC Input:** Inverter receives DC power from solar panels or batteries (e.g., 12V, 24V, or 48V DC)
2. **Electronic Switching:** Special electronic switches (transistors) rapidly turn on and off, chopping the DC into pulses
3. **Waveform Creation:** By controlling the timing of switches, the inverter creates an output that resembles AC sine wave
4. **Voltage Transformation:** Transformers increase the low DC voltage (12V, 24V) to higher AC voltage (220-240V)
5. **Filtering:** Filters smooth the switched output into clean AC power
6. **AC Output:** Final output is 220-240V AC at 50Hz, suitable for home appliances

Key Specifications:

Input Voltage: Must match your battery system

- 12V system: For small loads (up to 1kW)
- 24V system: For medium loads (1-3kW)
- 48V system: For large loads (3kW+)

Output Voltage: Standard in Nigeria is 220-240V AC

Frequency: 50Hz in Nigeria (60Hz in USA)

Power Rating:

- **Continuous Power:** What inverter supplies constantly (e.g., 2000W)
- **Surge Power:** Short-term power for starting motors (usually 2x continuous)

Efficiency: Good inverters are 85-95% efficient (some power lost as heat)

4. Care and Maintenance of Inverters

Installation Location:

- Install in cool, dry, well-ventilated area
- Keep away from direct sunlight and rain
- Maintain clearance around inverter for airflow (at least 15cm all sides)

- Mount on stable, vibration-free surface
- Keep away from flammable materials

Operating Precautions:

- Don't overload - stay within rated capacity
- Use proper cable sizes for connections
- Ensure good ventilation - inverters produce heat
- Keep away from water and moisture
- Don't cover or block cooling vents
- Turn off during extended non-use

Regular Maintenance:

Weekly:

- Check display for error messages
- Listen for unusual sounds (buzzing, clicking)
- Observe cooling fan operation

Monthly:

- Clean dust from cooling vents using soft brush or compressed air
- Check connections are tight
- Inspect cables for damage
- Check for signs of overheating

Quarterly:

- Test system performance
- Verify correct voltage output with multimeter
- Check battery connections
- Professional inspection recommended

Common Problems and Solutions:

Problem	Possible Cause	Solution
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Inverter shuts down	Overload	Reduce load, check ratings
Low battery alarm	Batteries discharged	Charge batteries, check solar production
Overheating alarm	Poor ventilation	Improve airflow, clean vents
No output	Blown fuse	Check and replace fuses
Humming noise	Overload or fault	Reduce load, check connections

Things to AVOID:

- ✗ Operating beyond rated capacity
- ✗ Blocking cooling vents
- ✗ Exposing to water or moisture
- ✗ Opening inverter cover (high voltage inside - dangerous!)
- ✗ Using with incompatible batteries
- ✗ Ignoring warning alarms

When to Call a Professional:

- Persistent error messages
- Strange burning smell
- Visible damage or burnt components
- Inverter won't turn on
- Continuous overheating
- Any internal component failure

EVALUATION

1. What is an inverter and why is it necessary in a solar system?
2. Explain the difference between DC and AC electricity.
3. Compare pure sine wave and modified sine wave inverters. Which is better for all appliances?
4. What is a hybrid inverter?
5. List the steps in how an inverter converts DC to AC power.

6. State FOUR safety precautions for inverter installation.
7. Describe THREE maintenance tasks for keeping inverters in good condition.
8. What should you do if your inverter frequently shuts down?

ASSIGNMENT

1. Draw a diagram showing how an inverter connects between batteries and home appliances.
2. Visit a home or shop with solar installation. Identify the inverter, note its specifications (power rating, input voltage), and write a brief report.
3. Create a troubleshooting guide for inverters with at least 5 common problems and their solutions.

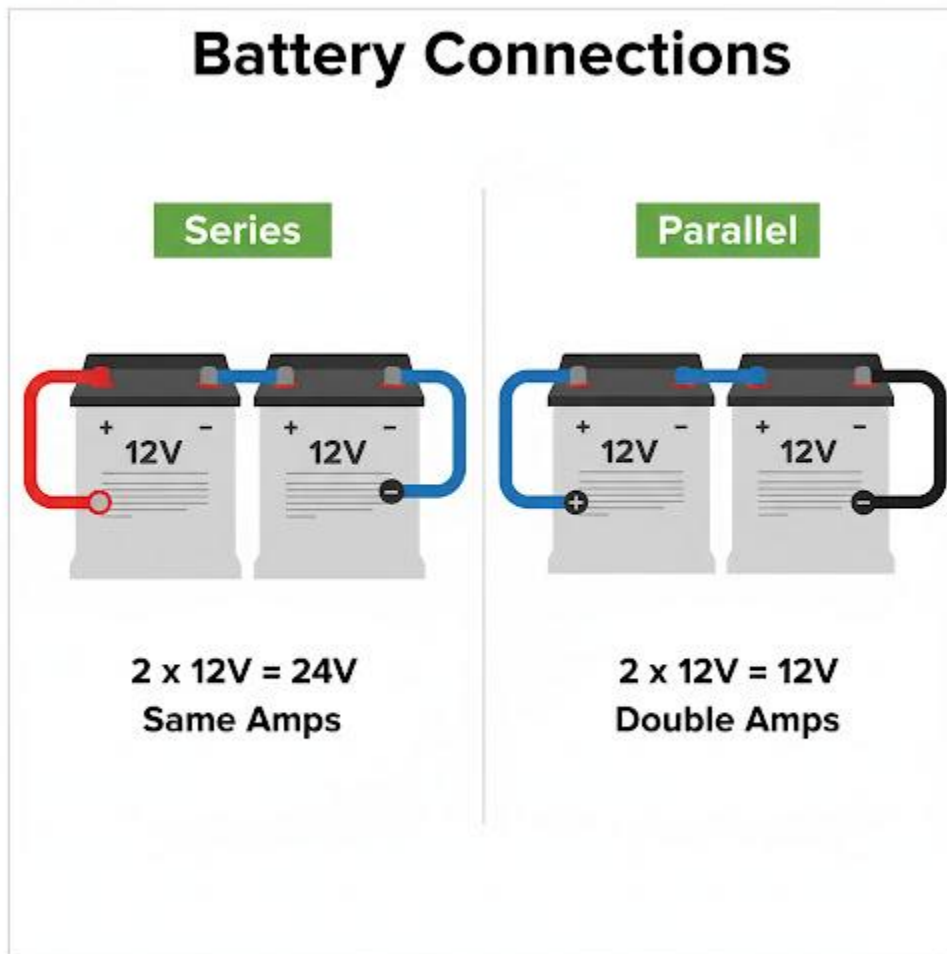


WEEK 9: SOLAR PV COMPONENTS - BATTERY BANK STORAGE

1. Definition and Importance of Battery Storage

Definition: A battery bank is a group of batteries connected together to store electrical energy produced by solar panels for use when the sun is not shining.

Here is a clear, labeled diagram illustrating 'Battery Connections':



Importance:

- Stores excess energy generated during the day
- Provides power at night and on cloudy days
- Ensures continuous power supply
- Acts as backup during low solar production

- Stabilizes system voltage

How It Works:

- **During daytime:** Solar panels charge batteries while powering appliances
- **At night:** Batteries supply power to appliances through the inverter
- The charge controller manages charging and prevents overcharging

2. Types of Solar Batteries

A) LEAD-ACID BATTERIES

Most common type for solar systems in Nigeria.

i) Flooded Lead-Acid Batteries:

- Contains liquid sulfuric acid
- **Advantages:** Least expensive, reliable, recyclable, locally available
- **Disadvantages:** Requires regular maintenance (water topping), produces hydrogen gas (needs ventilation), can spill acid, shorter lifespan (3-5 years)
- **Best For:** Budget systems, areas with maintenance capability

ii) Sealed/Maintenance-Free Lead-Acid Batteries:

- Sealed, no liquid acid to maintain
- Types: AGM (Absorbed Glass Mat) and Gel batteries
- **Advantages:** No maintenance required, can't spill, can be mounted in any position, less gassing
- **Disadvantages:** More expensive than flooded, sensitive to overcharging, must use correct charging settings
- **Best For:** Systems where maintenance is difficult

B) LITHIUM-ION BATTERIES

Modern technology increasingly used in solar systems.

Advantages:

- Longest lifespan (10-15 years)
- Deepest discharge allowed (80-90% depth)

- Lightweight (1/3 weight of lead-acid)
- Compact size
- No maintenance
- Built-in battery management
- Fast charging

Disadvantages:

- Very expensive (3-4x cost of lead-acid)
- Requires special charger
- Fire risk if damaged
- Not yet widely available in Nigeria

Best For: Long-term investment, weight-sensitive applications

Comparison Table:

Feature	Flooded Lead-Acid	Sealed Lead-Acid	Lithium-Ion
Cost	Lowest	Moderate	Highest
Lifespan	3-5 years	4-7 years	10-15 years
Maintenance	High	None	None
Depth of Discharge	50%	50%	80-90%
Weight	Heavy	Heavy	Light
Availability (Nigeria)	Excellent	Good	Limited

3. Battery Capacity and Voltage Rating

A) VOLTAGE RATING

Standard Battery Voltages:

- 12V: Most common for small to medium systems
- 24V: Medium to large systems
- 48V: Large systems (more efficient)

System Voltage Selection:

- Small systems (up to 1kW): 12V
- Medium systems (1-3kW): 24V
- Large systems (3kW+): 48V

Creating Battery Banks:

- **Series Connection:** Adds voltages ($2 \times 12V = 24V$, $4 \times 12V = 48V$)
- **Parallel Connection:** Adds capacity, keeps voltage same
- **Series-Parallel:** Combines both to achieve desired voltage and capacity

B) CAPACITY RATING

Measured in **Ampere-hours (Ah)**.

What Ah Means:

- 100Ah battery can supply:
 - 100A for 1 hour, OR
 - 50A for 2 hours, OR
 - 10A for 10 hours

Calculating Battery Needs:

Formula:

Battery Capacity (Ah) = (Daily Energy Consumption (Wh) × Days of Autonomy) ÷ (System Voltage × Depth of Discharge)

Example:

- Daily consumption: 3000Wh
- Days of backup desired: 2 days
- System voltage: 24V
- Depth of discharge: 50% (for lead-acid)

Battery Capacity = $(3000Wh \times 2 \text{ days}) \div (24V \times 0.5) = \mathbf{500Ah}$

You would need 500Ah at 24V (example: four 12V 250Ah batteries in series-parallel)

C Rating (Discharge Rate):

- **C10:** Capacity at 10-hour discharge rate (most common rating)
- **C20:** Capacity at 20-hour discharge rate

- Higher discharge rates reduce available capacity

4. Charging and Maintenance Tips

Proper Charging

Charging Stages:

1. **Bulk Charge:** Fast charging to 80%
2. **Absorption:** Slower charging to 100%
3. **Float:** Maintains full charge without overcharging

Charging Voltage (for 12V lead-acid):

- Bulk: 14.4-14.6V
- Absorption: 14.2-14.4V
- Float: 13.2-13.8V

Charging Best Practices:

- Use correct charge controller settings
- Never overcharge (damages batteries)
- Charge at moderate temperatures (20-25°C)
- Regular charging maintains battery health
- Avoid leaving batteries discharged for extended periods

Maintenance for Flooded Lead-Acid Batteries

Monthly:

- Check electrolyte levels
- Add distilled water if below plates (NEVER tap water)
- Never add acid unless instructed by professional
- Clean battery tops (dirt conducts current)
- Check for corrosion on terminals
- Verify voltage of each battery

Cleaning Corroded Terminals:

1. Disconnect battery
2. Mix baking soda with water
3. Apply to corroded areas with brush
4. Rinse with water
5. Dry thoroughly
6. Apply petroleum jelly to terminals
7. Reconnect

Quarterly:

- Check specific gravity with hydrometer (indicates charge level)
- Equalization charge (controlled overcharge to balance cells)
- Tighten loose terminals

General Maintenance Tips**Battery Life Extension Tips:**

- Avoid deep discharges (don't drain below 50% for lead-acid)
- Keep batteries fully charged when possible
- Avoid extreme temperatures
- Don't mix old and new batteries
- Use batteries of same type, age, and capacity together
- Follow manufacturer recommendations
- Keep batteries cool (high temperature shortens life)
- Store in well-ventilated area
- Check connections regularly
- Keep records of maintenance

Signs of Battery Problems:

- Rapid voltage drop under load
- Won't hold charge

- Swollen or deformed casing
- Acid leakage
- Excessive heat during charging
- Reduced backup time
- One battery significantly lower voltage than others

Safety Reminders:

- ⚠ Batteries produce explosive hydrogen gas
- Always work in ventilated areas
- No smoking or sparks near batteries
- Wear protective equipment
- Acid is extremely corrosive
- Disconnect before maintenance

EVALUATION

1. What is a battery bank and why is it important in solar systems?
2. Compare flooded lead-acid and lithium-ion batteries. State THREE advantages of each.
3. What does "Ah" mean? If a 200Ah battery supplies 50A, how long will it last?
4. Explain how to create a 24V battery bank using 12V batteries.
5. What is "depth of discharge" and why is it important?
6. List FIVE maintenance tasks for flooded lead-acid batteries.
7. Describe how to clean corroded battery terminals.
8. What are THREE signs that a battery is failing?

ASSIGNMENT

1. Calculate the battery capacity needed for a home that uses 4000Wh daily, requires 2 days backup, uses a 24V system with 50% depth of discharge. Show your working.
2. Draw diagrams showing: (a) Two 12V batteries connected in series (b) Two 12V batteries connected in parallel. Label the voltages clearly.

3. Create a monthly maintenance checklist for solar batteries.



WEEK 10: SYSTEM SIZING (BATTERY & INVERTER)

1. Meaning of System Sizing

Definition: System sizing is the process of calculating the correct capacity of solar system components (solar panels, batteries, inverter, charge controller) to meet specific energy needs.

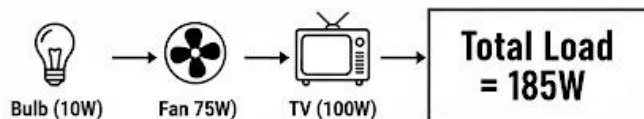
Why Proper Sizing Matters:

- **Too Small:** System can't power all appliances, frequent power shortages
- **Too Large:** Waste of money on unnecessary capacity
- **Proper Size:** Meets all needs efficiently at optimal cost

Steps in System Sizing:

1. Calculate total load (power needed)
2. Calculate daily energy consumption
3. Size inverter for maximum load
4. Size battery bank for storage needs
5. Size solar panels for daily energy and battery charging

SOLAR PV COMPONENTS SYSTEM SIZING HOW TO SIZE YOUR SYSTEM



↓

$$185W \times 6 \text{ hours} = 1110Wh \text{ Daily Energy}$$

2. Calculating Inverter Capacity

The inverter must handle your peak load plus surge power for motor-driven appliances.

Step-by-Step Process:

STEP 1: List All Appliances Identify all devices you want to power

STEP 2: Calculate Total Load Add wattage of all appliances that will run simultaneously

STEP 3: Add Safety Margin Multiply by 1.25 (25% safety margin)

STEP 4: Consider Surge Power Motor appliances (fans, refrigerators, pumps) need 2-3x their running power to start

Example:

Appliances to run simultaneously:

- 6 LED bulbs @ 10W = 60W
- 3 Fans @ 75W = 225W
- 1 TV @ 100W = 100W
- 1 Decoder @ 20W = 20W
- 1 Laptop @ 65W = 65W
- 1 Refrigerator @ 150W = 150W

Total Running Load = $60 + 225 + 100 + 20 + 65 + 150 = 620W$

With 25% safety margin = $620W \times 1.25 = 775W$

Surge Consideration:

- Refrigerator needs 2x for starting = $150W \times 2 = 300W$ surge
- Fans need 1.5x for starting = $225W \times 1.5 = 337.5W$ surge

Total Surge Load = $60 + 337.5 + 100 + 20 + 65 + 300 = 882.5W$

Recommended Inverter:

- Continuous Power: 800W to 1000W
- Surge Power: 1600W to 2000W
- **Choose: 1000W (1kW) inverter with 2000W surge capacity**

3. Calculating Battery Capacity

Batteries must store enough energy for nighttime use and backup days.

Formula:

Battery Capacity (Ah) = (Daily Energy (Wh) × Days of Autonomy) ÷ (System Voltage × Depth of Discharge × Inverter Efficiency)

Parameters:

- **Days of Autonomy:** Backup days needed (typically 1-3 days)
- **Depth of Discharge (DoD):**
 - Lead-acid: 50% (0.5)
 - Lithium-ion: 80-90% (0.8-0.9)
- **Inverter Efficiency:** Typically 85-90% (0.85-0.90)

Example:

Given:

- Daily energy consumption: 4000Wh
- Days of autonomy: 2 days
- System voltage: 24V
- Battery type: Lead-acid (50% DoD)
- Inverter efficiency: 90%

Calculation:

Battery Capacity = (4000Wh × 2) ÷ (24V × 0.5 × 0.9)

= 8000 ÷ 10.8

= 741Ah (round up to 750Ah or 800Ah)

Battery Configuration for 24V system at 750Ah:

- Four 12V 200Ah batteries properly configured in series-parallel

4. Matching Solar Components to Load Requirements

All components must work together efficiently.

Component Matching Checklist:

Voltage Compatibility:

- All batteries must be same voltage
- Charge controller voltage must match battery bank
- Inverter input must match battery bank voltage
- Solar panel array voltage must be compatible with charge controller

Current Capacity:

- Cables must handle maximum current
- Charge controller must handle solar panel current
- Inverter must handle maximum load current

Power Balance:

- Solar panels must generate enough daily energy
- Battery must store enough for nighttime/backup
- Inverter must handle peak loads

Example System Design:

Load Requirements:

- Daily energy: 3500Wh
- Peak load: 800W

System Design:

- **System Voltage:** 24V (appropriate for this size)
- **Inverter:** 1000W/2000W surge, 24V input
- **Battery Bank:** 600Ah at 24V (four 12V 300Ah batteries in series-parallel)
- **Solar Panels:** 1200W array (to generate 4000-5000Wh daily)
- **Charge Controller:** 50A MPPT, 24V system

5. Practical Exercise on Load Sizing

Exercise 1: Simple Home

Appliances:

- 4 LED bulbs (10W each) - 6 hours/day

- 2 fans (75W each) - 8 hours/day
- 1 TV (100W) - 4 hours/day
- 1 small fridge (120W) - 12 hours/day (cycling)

Tasks: a) Calculate total daily energy consumption b) Calculate peak load (all on simultaneously)
c) Size appropriate inverter d) Size battery bank for 2 days backup (24V system, lead-acid)

Solution:

a) Daily Energy:

- Bulbs: $40\text{W} \times 6\text{h} = 240\text{Wh}$
- Fans: $150\text{W} \times 8\text{h} = 1200\text{Wh}$
- TV: $100\text{W} \times 4\text{h} = 400\text{Wh}$
- Fridge: $120\text{W} \times 12\text{h} = 1440\text{Wh}$
- **Total = 3280Wh $\approx$ 3.3kWh**

b) Peak Load: $40 + 150 + 100 + 120 = 410\text{W}$

c) Inverter:

- With safety margin: $410\text{W} \times 1.25 = 512.5\text{W}$
- Considering surge (fridge 2x, fans 1.5x): $\sim 750\text{W}$
- **Recommended: 800W/1600W surge inverter**

d) Battery:

$$(3280\text{Wh} \times 2) \div (24\text{V} \times 0.5 \times 0.9) = 607\text{Ah}$$

- **Use: Four 12V 200Ah batteries (800Ah at 24V)**

EVALUATION

1. What is system sizing and why is it important?
2. List the main steps in sizing a solar PV system.
3. Calculate the inverter size needed for: 5 bulbs (12W), 4 fans (70W), 2 TVs (100W each), 1 laptop (60W) running simultaneously.
4. Explain what "depth of discharge" means and give typical values for lead-acid and lithium-ion batteries.

5. A home uses 5000Wh daily. Calculate battery capacity for 2 days backup, 24V system, lead-acid batteries (50% DoD), 90% inverter efficiency.
6. Why is a 25% safety margin added when sizing inverters?
7. What factors must be considered when matching system components?

ASSIGNMENT

1. **Design a complete solar system for your home:**
 - List all appliances with power ratings and hours of use
 - Calculate daily energy consumption
 - Calculate peak load
 - Size inverter
 - Size battery bank (assume 24V system, 2 days backup, lead-acid)
 - Draw a simple diagram showing your system layout
 - Present in a neat table format
2. **Compare two system designs:**
 - System A: 12V, 1000W inverter, 400Ah batteries
 - System B: 24V, 1000W inverter, 200Ah batteries
 - Explain which is better and why
3. **Calculate how many 200W solar panels are needed to generate 4500Wh per day** (assume 5 peak sun hours). Show your working.

WEEK 11: REVISION

Comprehensive Review of First Term Topics

Topics Covered This Term:

1. Introduction to Solar PV Systems (Week 1)

- Meaning and importance of solar energy
- Comparison with other energy sources
- Components of PV systems
- Advantages in homes and schools

2. Power Load and Load Sizing (Week 2)

- Understanding electrical load
- Appliance power ratings
- Calculating total load
- Energy consumption calculations

3. Electrical Safety (Week 3)

- Electrical safety definition
- Causes of electrical accidents
- Safety rules
- First aid for electric shock

4. Safety in Handling PV Materials (Week 4)

- Safe handling of panels, batteries, cables
- Tool storage and maintenance
- Personal protective equipment
- Environmental considerations

5. Solar Panels (Week 6)

- Function of solar panels
- Types: Monocrystalline, Polycrystalline, Thin Film
- Installation positioning
- Cleaning and maintenance

6. Inverters (Week 8)

- Inverter function and importance
- Types of inverters
- DC to AC conversion
- Inverter maintenance

7. Battery Storage (Week 9)

- Battery function and importance
- Types: Lead-acid vs Lithium-ion
- Capacity and voltage ratings
- Charging and maintenance

8. System Sizing (Week 10)

- Importance of proper sizing
- Inverter capacity calculation
- Battery capacity calculation
- Component matching

Key Formulas to Remember:

1. **Total Load = Sum of all appliances running simultaneously**
2. **Energy (Wh) = Power (W) × Time (hours)**
3. **Inverter Size = Total Load × 1.25 (safety margin)**
4. **Battery Capacity (Ah) = (Daily Energy × Days) ÷ (Voltage × DoD × Efficiency)**
5. **1 kW = 1000W**

**SECOND TERM - JSS 1 SOLAR PHOTOVOLTAIC (PV) INSTALLATION AND
MAINTENANCE LESSON NOTES**



WEEK 1: REVIEW OF FIRST TERM WORK

1. Recap of Solar Components and Their Functions

Solar Panels

- Convert sunlight directly into DC electricity
- **Types:** Monocrystalline (most efficient), Polycrystalline (moderate), Thin Film (least efficient)
- Should face south in Nigeria
- Require regular cleaning for optimal performance

Charge Controller

- Regulates power flow from panels to batteries
- Prevents overcharging and battery damage
- **Types:** PWM (cheaper) and MPPT (more efficient)

Battery Bank

- Stores electrical energy for nighttime and backup use
- **Types:** Lead-acid (most common, needs maintenance) and Lithium-ion (expensive, long-lasting)
- Capacity measured in Ampere-hours (Ah)
- Voltage ratings: 12V, 24V, or 48V

Inverter

- Converts DC to AC electricity
- **Types by waveform:** Pure sine wave (best), Modified sine wave (basic)
- **Types by application:** Stand-alone, Grid-tie, Hybrid
- Must be sized to handle peak load plus surge power

Cables and Connectors

- Connect all system components
- Must be properly sized for current
- Should be protected in conduits

Mounting Structure

- Holds panels securely in position

- Must be strong and properly angled

2. Review of Load Sizing and Safety Precautions

Load Sizing Review

Key Formulas:

- Total Load (W) = Sum of all appliances running simultaneously
- Energy (Wh) = Power (W) × Time (hours)
- Inverter Size = Total Load × 1.25 (safety margin)
- Battery Capacity (Ah) = (Daily Energy × Days) ÷ (Voltage × DoD × Efficiency)

Practice Problem:

Calculate system requirements for:

- 5 bulbs (10W) - 6 hours/day
- 2 fans (75W) - 8 hours/day
- 1 TV (100W) - 4 hours/day
- 1 fridge (150W) - 12 hours/day

Solution:

- **Daily Energy:** $(50 \times 6) + (150 \times 8) + (100 \times 4) + (150 \times 12) = 300 + 1200 + 400 + 1800 = 3,700\text{Wh}$
- **Peak Load:** $50 + 150 + 100 + 150 = 450\text{W}$
- **Inverter:** $450\text{W} \times 1.25 = 562.5\text{W}$ (Use **800W inverter** considering surge)
- **Battery (24V, 2 days, 50% DoD):** $(3700 \times 2) \div (24 \times 0.5 \times 0.9) = 686\text{Ah}$ (Use **4×200Ah batteries**)

Safety Precautions Review

Electrical Safety:

- Never touch electrical devices with wet hands
- Don't overload sockets
- Switch off before unplugging

- Keep electrical items away from water
- Report damaged cables immediately

Solar Panel Safety:

- Always carry with two people
- Wear gloves and safety shoes
- Cover panels when working on connections
- Store properly to prevent damage

Battery Safety:

- Wear protective equipment (goggles, gloves, apron)
- Work in well-ventilated areas
- Never smoke near batteries
- Connect positive to positive, negative to negative
- If acid touches skin, wash immediately with water

Inverter Safety:

- Install in cool, dry, ventilated area
- Don't overload beyond capacity
- Keep cooling vents clean
- Don't open inverter cover

3. Correction of Common Errors in Handling PV Materials

Common Mistakes and Corrections:

Error 1: Incorrect Cable Sizing

- **Mistake:** Using thin cables for high current
- **Problem:** Cables overheat and can cause fire
- **Correction:** Always use properly sized cables for the current load

Error 2: Wrong Battery Connections

- **Mistake:** Mixing old and new batteries, or different capacities
- **Problem:** Uneven charging/discharging, reduced lifespan

- **Correction:** Use batteries of same type, age, and capacity

Error 3: Poor Panel Positioning

- **Mistake:** Installing panels facing wrong direction or with shade
- **Problem:** Significantly reduced power output
- **Correction:** Face panels south, ensure no shade throughout the day

Error 4: Overloading Inverter

- **Mistake:** Connecting more appliances than inverter capacity
- **Problem:** Inverter shuts down or gets damaged
- **Correction:** Calculate total load before connecting appliances

Error 5: Neglecting Battery Maintenance

- **Mistake:** Not checking water levels in flooded batteries
- **Problem:** Battery damage and shortened lifespan
- **Correction:** Check and top up with distilled water monthly

Error 6: Improper Cleaning of Panels

- **Mistake:** Cleaning hot panels with cold water
- **Problem:** Thermal shock can crack glass
- **Correction:** Clean early morning or evening with appropriate materials

Error 7: Ignoring System Monitoring

- **Mistake:** Not checking system performance regularly
- **Problem:** Problems go unnoticed until major failure
- **Correction:** Monitor displays, check voltages, record performance

Error 8: Poor Ventilation for Batteries

- **Mistake:** Storing batteries in enclosed, unventilated space
- **Problem:** Hydrogen gas accumulation can cause explosion
- **Correction:** Ensure adequate ventilation in battery storage area

Error 9: Wrong Depth of Discharge

- **Mistake:** Draining lead-acid batteries below 50%
- **Problem:** Significantly shortens battery life

- **Correction:** Use appropriate DoD for battery type

Error 10: Not Using PPE

- **Mistake:** Working without safety equipment
- **Problem:** Risk of injury from accidents
- **Correction:** Always wear appropriate PPE for the task

EVALUATION

1. Name and explain the function of any FOUR components of a solar PV system.
2. Calculate the daily energy consumption for: 4 bulbs (15W, 8hrs), 3 fans (70W, 10hrs), 1 TV (120W, 5hrs).
3. What inverter size is needed for a total load of 600W?
4. List SIX electrical safety rules.
5. State FOUR common mistakes in handling solar PV materials and how to correct them.
6. Why should batteries be installed in well-ventilated areas?
7. Calculate battery capacity for 4000Wh daily consumption, 2 days backup, 24V system, 50% DoD.

ASSIGNMENT

1. **Create a comprehensive summary chart showing:**
 - Each solar component
 - Its function
 - Common problems
 - Safety precautions
2. **Design three practice problems on load sizing and solve them completely.**
3. **Write a one-page reflection** on the most important safety lesson you learned in first term and why it matters.

WEEK 2: CONCEPT OF ENTREPRENEURSHIP

Meaning of Entrepreneurship

Definition: Entrepreneurship is the process of starting and running a business venture by identifying opportunities, taking calculated risks, and organizing resources to create value and make profit.

Key Terms:

- **Entrepreneur:** A person who starts and operates a business, taking financial risks with the hope of making profit
- **Enterprise:** A business organization or project
- **Innovation:** Creating new ideas, products, or methods to solve problems or meet needs

Characteristics of Entrepreneurship:

- Involves identifying business opportunities
- Requires planning and organization
- Involves taking calculated risks
- Aims to create value and generate profit
- Requires hard work, dedication, and persistence

Examples of Entrepreneurs:

- A person who starts a solar installation business
- Someone who sells solar products (panels, batteries, inverters)
- A technician who offers solar maintenance services
- A person who manufactures solar mounting structures

Importance of Entrepreneurship in Renewable Energy

Economic Benefits

- **Job Creation:** Solar businesses employ installers, technicians, salespeople, and support staff
- **Income Generation:** Entrepreneurs earn profits from their businesses
- **Economic Growth:** Small businesses contribute to national GDP
- **Wealth Creation:** Successful businesses create wealth for entrepreneurs and communities

Energy Security

- Reduces dependence on unreliable grid electricity
- Provides alternative power solutions
- Makes clean energy accessible to more people
- Supports rural electrification

Environmental Benefits

- Promotes clean energy adoption
- Reduces carbon emissions
- Decreases fossil fuel consumption
- Creates awareness about sustainable living

Social Benefits

- Improves quality of life through reliable electricity
- Enables better education (powered schools)
- Supports healthcare (powered clinics)
- Empowers communities with energy access

Technology Advancement

- Encourages innovation in solar technology
- Improves installation techniques
- Develops better maintenance practices
- Creates local technical expertise

Skills Development

- Provides training opportunities
- Develops technical and business skills
- Creates career pathways for youth
- Transfers knowledge within communities

Why Solar Energy Entrepreneurship is Important in Nigeria:

- Unstable PHCN supply creates huge market demand

- Growing awareness of renewable energy benefits
- Government support for renewable energy initiatives
- Decreasing costs of solar components
- Rising fuel prices make solar more competitive
- Large population without electricity access

Characteristics of a Good Entrepreneur

A) PERSONAL QUALITIES

1. Self-Confidence

- Believes in their abilities
- Not easily discouraged by challenges
- Makes decisions with conviction
- Inspires confidence in customers and employees

2. Hard Work and Dedication

- Willing to work long hours
- Puts in effort to ensure business success
- Doesn't give up when facing difficulties
- Persists until goals are achieved

3. Risk-Taking Ability

- Willing to take calculated risks
- Not afraid of failure
- Learns from mistakes
- Makes bold decisions when necessary

4. Creativity and Innovation

- Thinks of new solutions to problems
- Improves existing products or services
- Adapts to changing market needs
- Finds unique ways to serve customers

5. Self-Discipline

- Manages time effectively
- Keeps promises and commitments
- Maintains focus on goals
- Controls personal spending

6. Honesty and Integrity

- Deals fairly with customers
- Delivers quality products/services
- Honors agreements
- Builds trust through ethical behavior

B) BUSINESS SKILLS

7. Planning Ability

- Sets clear business goals
- Develops strategies to achieve objectives
- Anticipates future needs and challenges
- Prepares for different scenarios

8. Good Communication

- Explains products/services clearly
- Listens to customer needs
- Negotiates effectively
- Maintains good relationships

9. Financial Management

- Keeps accurate records
- Controls costs
- Manages cash flow
- Makes sound financial decisions

10. Problem-Solving Skills

- Identifies problems quickly
- Analyzes situations carefully
- Develops practical solutions
- Implements solutions effectively

11. Customer Service Orientation

- Puts customer satisfaction first
- Responds promptly to complaints
- Goes extra mile to help customers
- Builds long-term relationships

12. Continuous Learning

- Stays updated on industry developments
- Seeks new knowledge and skills
- Learns from competitors and mentors
- Adapts to technological changes

Entrepreneurial Opportunities in Solar Energy

A) SOLAR INSTALLATION SERVICES

- Residential installations (homes)
- Commercial installations (offices, shops)
- Industrial installations (factories)
- Agricultural installations (farms, irrigation)
- Street light installations

Requirements: Technical skills, tools, business registration

B) SOLAR PRODUCT SALES

- Selling solar panels
- Batteries and inverters
- Charge controllers

- Solar lights and lanterns
- Accessories (cables, connectors, mounting hardware)

Requirements: Capital for stock, storage space, sales skills

C) SOLAR MAINTENANCE SERVICES

- Regular system cleaning and inspection
- Battery maintenance and replacement
- Troubleshooting and repairs
- System upgrades
- Performance monitoring

Requirements: Technical knowledge, testing equipment, service vehicle

D) SOLAR TRAINING AND CONSULTING

- Training technicians and installers
- Providing system design consultation
- Energy audits for homes and businesses
- Technical advisory services

Requirements: Expertise, training materials, good communication skills

E) SOLAR COMPONENT MANUFACTURING

- Making mounting structures
- Assembling battery banks
- Producing solar accessories
- Manufacturing solar-powered devices

Requirements: Workshop space, tools, technical skills, capital

F) SOLAR SYSTEM MONITORING

- Offering remote monitoring services

- System performance analysis
- Maintenance scheduling
- Efficiency optimization

Requirements: Monitoring software, technical knowledge

G) SPECIALIZED SERVICES

- Solar water pumping systems
- Solar refrigeration for businesses
- Solar-powered security systems
- Mobile charging stations
- Solar kiosks for rural areas

EVALUATION

1. Define entrepreneurship.
2. Explain FOUR reasons why entrepreneurship is important in renewable energy.
3. List and explain SIX characteristics of a good entrepreneur.
4. Mention FIVE entrepreneurial opportunities in solar energy business.
5. Why is risk-taking an important quality for entrepreneurs?
6. How does solar entrepreneurship contribute to environmental protection?
7. What small solar business can a student start?

ASSIGNMENT

1. **Interview a solar business owner in your area.** Write a one-page report covering:
 - Type of business they run
 - How they started
 - Challenges they face
 - Advice for young people interested in solar business
2. **List TEN entrepreneurial characteristics and rate yourself honestly on each (1-10).**
Write one paragraph on which characteristics you need to develop and how.

3. **Research and write about TWO successful Nigerian entrepreneurs in renewable energy.**



WEEK 3: CONCEPT OF MONEY

Meaning and Functions of Money

Definition: Money is anything generally accepted as a medium of exchange for goods and services and for payment of debts.

Forms of Money:

- **Coins:** Metal money (₦1, ₦2)
- **Notes:** Paper money (₦5 to ₦1000)
- **Electronic Money:** Bank transfers, mobile money, online payments
- **Cheques:** Written orders to banks to pay specific amounts

Functions of Money

1. Medium of Exchange

- Used to buy goods and services
- Eliminates barter system
- Makes trade easier and faster

2. Measure of Value (Unit of Account)

- Provides common measure for pricing
- Makes value comparison easy
- Enables record keeping

3. Store of Value

- Can be saved for future use
- Preserves wealth over time
- Allows capital accumulation

4. Standard of Deferred Payment

- Used to settle debts
- Enables credit transactions
- Allows payment installments

5. Transfer of Value

- Easily transferred between people
- Facilitates long-distance transactions
- Enables payment for services

Importance of Money in Business

For Starting a Business

1. Capital for Setup:

- Equipment and tools
- Initial stock
- Shop/office space
- Business registration fees

2. Operational Expenses:

- Raw materials and components
- Employee salaries
- Transportation costs
- Utilities and marketing

3. Growth and Expansion:

- Additional equipment
- Hiring more staff
- Opening new branches
- Expanding product range

4. Emergency Fund:

- Unexpected repairs
- Slow business periods
- Equipment breakdowns

For Solar Business Specifically

Initial Setup Costs:

- Solar installation tools: ₦100,000 - ₦500,000
- Initial component stock: ₦500,000 - ₦2,000,000
- Training and certification: ₦50,000 - ₦200,000
- Business registration: ₦20,000 - ₦50,000
- Shop rent (6 months): ₦300,000 - ₦600,000

Monthly Running Costs:

- Staff salaries: ₦60,000 - ₦150,000 per staff
- Transportation: ₦30,000 - ₦100,000
- Utilities: ₦10,000 - ₦30,000
- Marketing: ₦20,000 - ₦50,000
- Miscellaneous: ₦20,000 - ₦50,000

Sources of Business Capital

A) PERSONAL SOURCES

1. Personal Savings

- **Advantages:** No interest, full control, no debt
- **Disadvantages:** May be insufficient, takes time to accumulate

2. Family and Friends

- **Advantages:** Flexible terms, lower/no interest
- **Disadvantages:** Can strain relationships if business fails

3. Personal Assets

- **Advantages:** Generates capital quickly
- **Disadvantages:** Risk losing valuable assets

B) EXTERNAL SOURCES

4. Bank Loans

- **Advantages:** Large amounts available, professional relationship
- **Disadvantages:** Interest payments, strict requirements, collateral needed

5. Microfinance Banks

- **Advantages:** Lower collateral requirements, understands small business
- **Disadvantages:** Smaller amounts, higher interest rates

6. Cooperative Societies

- **Advantages:** Supportive community, reasonable terms
- **Disadvantages:** Limited amounts, must be member

7. Government Grants and Programs

- **Advantages:** Often interest-free or low interest
- **Disadvantages:** Competitive, complex application process

8. Business Partners/Investors

- **Advantages:** Shares risk, brings connections and experience
- **Disadvantages:** Shared profits, shared control

9. Trade Credit

- **Advantages:** Immediate stock access, pay after sales
- **Disadvantages:** Limited amounts, must build supplier trust

10. Business Competitions and Grants

- **Advantages:** Free money, no repayment
- **Disadvantages:** Highly competitive, time-consuming

Choosing the Right Source - Consider:

- Amount needed
- Urgency of need
- Ability to repay
- Interest rates and terms
- Risk tolerance
- Business stage

Role of Money Management in Entrepreneurship

Definition: Money management is the process of planning, organizing, controlling, and monitoring financial resources to achieve business goals.

Importance of Money Management

1. **Ensures Business Survival:** Prevents cash shortages, covers expenses
2. **Enables Profitability:** Controls costs, maximizes profit
3. **Facilitates Growth:** Provides funds for expansion
4. **Builds Credibility:** Maintains good credit, attracts investors
5. **Provides Decision-Making Information:** Shows profitability, guides pricing

Key Money Management Practices

A) BUDGETING Planning how money will be spent and allocated

Example Monthly Budget:

- Income: ₦500,000 (installations, sales, services)
- Expenses: Component purchases, salaries, transport, rent, marketing, utilities, savings, contingency = ₦500,000

B) RECORD KEEPING

- Write down all income and expenses
- Keep receipts and invoices
- Track money owed and owing

C) SEPARATING BUSINESS AND PERSONAL MONEY

- Use separate bank accounts
- Don't use business money for personal needs
- Pay yourself a regular salary

D) CONTROLLING COSTS

- Buy supplies at best prices
- Avoid unnecessary expenses
- Negotiate with suppliers
- Maintain equipment

E) PRICING STRATEGY Formula: Selling Price = Cost of Materials + Labor + Overhead + Desired Profit

Example:

- Materials: ₦120,000
- Labor: ₦40,000
- Overhead: ₦15,000
- Profit (30%): ₦52,500
- **Total: ₦227,500**

F) CASH FLOW MANAGEMENT

- Ensure enough cash when needed
- Time receipts and payments
- Plan for seasonal variations

G) DEBT MANAGEMENT

- Borrow only when necessary
- Understand loan terms
- Pay debts on time

H) SAVING AND INVESTMENT

- Set aside emergency funds
- Reinvest profits for growth
- Build capital reserves

Common Money Management Mistakes

1. Mixing business and personal money → **Use separate accounts**
2. No record keeping → **Document daily**
3. Spending all profits → **Save and reinvest**
4. Incorrect pricing → **Calculate costs carefully**
5. Poor credit control → **Collect payments promptly**
6. No emergency fund → **Save regularly**
7. Excessive borrowing → **Borrow only what you can repay**

EVALUATION

1. Define money and list THREE forms of money.
2. Explain FIVE functions of money with examples.
3. Why is money important for starting and running a business?
4. List and explain SIX sources of business capital.
5. What is the difference between a bank loan and a government grant?
6. Define money management and explain its importance in entrepreneurship.
7. State FIVE money management practices essential for business success.
8. Calculate the selling price for a solar installation with: Materials ₦150,000, Labor ₦50,000, Overhead ₦20,000, desired 25% profit.

ASSIGNMENT

1. **Create a detailed monthly budget for a small solar installation business.** Include at least 8 income sources and 10 expense categories. Ensure income exceeds expenses and include savings.
2. **Compare three sources of business capital** (personal savings, bank loan, business partner) by creating a table showing advantages, disadvantages, and best use cases for each.
3. **Write a one-page essay on "Why Money Management is Critical for Solar Business Success"** using examples from your community.

WEEK 4: BOOKKEEPING IN SOLAR PV INSTALLATION AND MAINTENANCE

1. Meaning of Bookkeeping

Definition: Bookkeeping is the systematic recording, organizing, and maintaining of financial transactions and records of a business.

Key Terms:

- **Transaction:** Business activity involving money
- **Record:** Written documentation of transactions
- **Entry:** Single recording of a transaction
- **Books:** Records where transactions are recorded

Purpose:

- Tracks money in and out
- Provides accurate financial information
- Helps make business decisions
- Required for tax purposes
- Shows profitability
- Identifies financial problems early

Bookkeeping vs. Accounting:

Bookkeeping	Accounting
Recording transactions	Interpreting financial records
Day-to-day documentation	Preparing financial reports
Basic clerical work	Requires professional training

2. Simple Record Keeping in Solar Business

A) BASIC BOOKS TO MAINTAIN

1. Cash Book: Records all cash received and paid out

DATE	DETAILS	CASH IN (₹)	CASH OUT (₹)	BALANCE (₹)
------	---------	-------------	--------------	-------------

2. Sales Book: Records all sales of products and services

DATE	CUSTOMER	PRODUCT/SERVICE	QTY	UNIT PRICE	TOTAL	RECEIPT NO.
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3. Purchase Book: Records all purchases

DATE	SUPPLIER	ITEMS	QTY	UNIT PRICE	TOTAL	INVOICE NO.
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4. Debtors Book (Receivables): Money owed to the business

DATE	CUSTOMER	AMOUNT OWED	AMOUNT PAID	BALANCE	DUE DATE
------	----------	-------------	-------------	---------	----------

5. Creditors Book (Payables): Money business owes suppliers

DATE	SUPPLIER	AMOUNT OWED	AMOUNT PAID	BALANCE	DUE DATE
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6. Inventory/Stock Book: Records stock of components

DATE	ITEM	OPENING STOCK	PURCHASED	SOLD/USED	CLOSING STOCK	VALUE
------	------	---------------	-----------	-----------	---------------	-------

B) SUPPORTING DOCUMENTS

1. Receipts: Issued when money received (numbered, include date, amount, customer, purpose)

2. Invoices: Sent to customer showing amount owed

3. Vouchers: Document for money paid out

4. Bank Statements: Monthly record from bank

3. Recording Income and Expenditure

INCOME (Money Received)

Sources in Solar Business:

- Sales of solar components
- Installation services
- Maintenance and repair services
- Consultation and training fees

- Commission from referrals

How to Record Income:

1. Issue Receipt
2. Record in Cash Book
3. Record in Sales Book
4. Update Stock Records
5. File Documents

Example Entry:

- Date: 15/03/2025
- Customer: Mrs. Adeyemi
- Service: 3kW solar system installation
- Amount: ₦550,000

EXPENDITURE (Money Paid Out)**Types of Expenses:****1. Direct Costs (Variable):**

- Components for specific jobs
- Transportation for installations
- Casual labor

2. Overhead Costs (Fixed):

- Shop rent
- Staff salaries
- Utilities
- Marketing
- Tools maintenance
- Insurance

How to Record Expenditure:

1. Obtain Receipt/Invoice

2. Prepare Payment Voucher
3. Record in Cash Book
4. Record in Purchase Book
5. Update Stock Records
6. File Documents

Calculating Profit

Formula: Profit = Total Income - Total Expenditure

Example:

- Total Income: ₦1,370,000
- Total Expenditure: ₦920,000
- **PROFIT = ₦450,000**

4. Preparing Simple Daily Sales Records

Purpose:

- Track daily performance
- Identify peak sales days
- Monitor cash flow
- Detect discrepancies quickly

Daily Sales Record Format

DATE: _____

TIM E	RECEIP T NO.	CUSTOME R	PRODUCT/SERVIC E	QT Y	AMOUN T (₦)	PAYMEN T
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DAILY SUMMARY:

- Total Cash Sales: ₦_____
- Total Transfer: ₦_____
- Part Payments: ₦_____

- **TOTAL SALES: ₦**_____
- Number of Customers: _____
- Number of Transactions: _____

End-of-Day Procedures

1. Count Cash
2. Verify Against Records
3. Reconcile Differences
4. Prepare Bank Deposit
5. Update Main Books
6. Secure Documents
7. Report Summary

Weekly Sales Summary

WEEK: _____

Day	Installations	Products	Services	Total Sales
Mon				
Tue				
WEEKLY TOTAL:				

Best Practices for Bookkeeping

1. **Record Immediately:** Don't delay documentation
2. **Be Accurate:** Double-check all figures
3. **Keep All Documents:** File receipts and invoices
4. **Review Regularly:** Check books daily
5. **Back Up Records:** Make copies
6. **Use Consistent Format:** Maintain standards
7. **Separate Business and Personal:** Never mix

8. **Secure Records:** Store safely
9. **Train Staff:** Ensure everyone follows procedures
10. **Consider Software:** Use accounting apps (QuickBooks, Excel)

EVALUATION

1. Define bookkeeping and explain its importance in solar business.
2. List and explain FIVE basic books that should be maintained in a solar business.
3. What information should be included in a receipt?
4. Differentiate between debtors and creditors.
5. Explain the proper procedure for recording income in a solar business.
6. Calculate profit if total income is ₦850,000 and total expenditure is ₦620,000.
7. Why is it important to prepare daily sales records?
8. Design a simple cash book format with appropriate columns.

ASSIGNMENT

1. **Create a complete set of business records for a fictional solar business for one week.**
Include:
 - Cash book with at least 10 transactions
 - Sales book with at least 5 sales
 - Purchase book with at least 4 purchases
 - Daily sales records for 5 days
 - Weekly summary
2. **Design professional templates for:**
 - Receipt
 - Invoice
 - Payment voucher

Include all necessary information fields.

3. **Visit a small solar business (or any business) in your area.** Ask about their bookkeeping system and write a one-page report on what you learned.

WEEK 6: BUSINESS START-UP IN SOLAR PV

1. Steps in Starting a Small Solar Business

STEP 1: DEVELOP YOUR IDEA

- Identify niche (installations, sales, maintenance)
- Research market, customers, competitors

STEP 2: ACQUIRE SKILLS

- Technical: Installation, maintenance, troubleshooting
- Business: Management, bookkeeping, marketing

STEP 3: WRITE A BUSINESS PLAN Include: Business name/location, services, target customers, market analysis, financial projections

Example: BrightLight Solar - Ikeja, Lagos | Startup: ₦800,000 | Monthly Profit: ₦150,000

STEP 4: SECURE CAPITAL Sources: Personal savings, family/friends, bank loans, government programs, microfinance

STEP 5: REGISTER BUSINESS

- Register with CAC
- Obtain TIN
- Register with local government

STEP 6: GET TOOLS & EQUIPMENT Tools (₦70,000-₦231,000): Multimeter, wire tools, drill, safety equipment, ladder Stock: Panels, batteries, inverters, cables, accessories

STEP 7: SET UP LOCATION Options: Home-based, shared space, rented shop, mobile service

STEP 8: BUILD SUPPLIER NETWORK Identify reliable suppliers, negotiate terms and discounts

STEP 9: HIRE STAFF (if needed) Start small, hire as business grows

STEP 10: LAUNCH Announce opening, offer promotions, deliver excellent service

2. Identifying Customer Needs

Methods:

- **Direct Conversations:** Ask about challenges, budget, appliances
- **Observation:** Visit locations, assess power situation

- **Surveys:** Reach many customers
- **Competitor Analysis:** Study offerings, identify gaps

Common Needs:

1. Reliability (consistent systems)
2. Affordability (reasonable prices, payment plans)
3. Quality (durable, professional installation)
4. Technical Support (after-sales, maintenance)
5. Energy Independence (no PHCN/generator dependence)
6. Correct Sizing (powers all appliances)
7. Professional Service (knowledgeable, clean work)

3. Marketing Solar Products/Services

The 4 Ps:

1. **Product:** What you sell (quality, features)
2. **Price:** How much you charge
3. **Place:** Where customers find you
4. **Promotion:** How you communicate

Marketing Strategies:

- **Word of Mouth:** Excellent service, referral incentives
- **Local Advertising:** Flyers, posters, signboards, business cards
- **Demonstrations:** Show working systems
- **Social Media:** Facebook, WhatsApp Business
- **Partnerships:** Electricians, contractors, agents
- **Community Engagement:** Sponsor events, give talks
- **Vehicle Branding:** Business name on vehicle
- **Promotions:** Discounts, free assessments

4. Business Ethics & Customer Relations

Business Ethics

Key Principles:

1. **Honesty:** Tell truth, don't exaggerate
2. **Integrity:** Keep promises, use quality materials
3. **Fairness:** Reasonable prices, equal treatment
4. **Responsibility:** Fix mistakes, provide support
5. **Transparency:** Clear costs, no hidden fees
6. **Respect:** Value time, listen, maintain confidentiality

Avoid: Substandard products, overcharging, false promises, abandoning jobs

Customer Relations Best Practices:

1. **Professional Communication:** Respond promptly, speak politely
2. **Understand Needs:** Ask questions, visit site, explain options
3. **Clear Agreements:** Written quotations, realistic timelines
4. **Quality Delivery:** Arrive on time, work professionally, test thoroughly
5. **After-Sales:** Follow up, provide training, respond to complaints
6. **Handle Complaints:** Listen, apologize, fix promptly, follow up
7. **Build Relationships:** Remember customers, offer loyalty benefits

EVALUATION

1. List FIVE main steps in starting a solar business.
2. What is a business plan and why is it important?
3. Explain FOUR methods of identifying customer needs.
4. What are the 4 Ps of marketing?
5. List SIX marketing strategies for solar business.
6. Define business ethics and explain its importance.
7. Describe FIVE customer relations best practices.
8. How should you handle customer complaints?

ASSIGNMENT

1. **Write a simple business plan** (2-3 pages): name, services, target customers, startup capital, income/expenses.
2. **Design a marketing flyer** for solar installation business.
3. **Create a "Customer Service Guidelines" poster** with 10 do's and don'ts.



WEEK 8: MARKETING IN SOLAR PV

1. Meaning of Marketing

Definition: Identifying, anticipating, and satisfying customer needs profitably through communication, promotion, and delivery.

Marketing vs. Selling:

- **Marketing:** Customer-focused, creates demand, builds long-term relationships
- **Selling:** Product-focused, fulfills demand, short-term transactions

Marketing Process: Research → Product Development → Pricing → Promotion → Distribution → Sales → After-Sales

2. Types of Marketing

A) MARKETING SOLAR GOODS (Products)

Examples: Panels, batteries, inverters, controllers, cables, lights

Strategies:

1. **Display:** Attractive showroom, working models, clear labeling
2. **Packaging:** Professional, warranty details, installation guides
3. **Differentiation:** Unique features, better warranties, bundle deals
4. **Pricing:** Competitive, penetration, bundle, psychological (₦99,500 vs ₦100,000)
5. **Promotion:** Demonstrate quality, show certifications, offer trials

B) MARKETING SOLAR SERVICES

Examples: Installation, maintenance, repairs, upgrades, energy audits

Strategies:

1. **Quality:** Professional workmanship, certified technicians, clean work
2. **Reliability:** Punctuality, consistent quality, honoring commitments
3. **Differentiation:** Specialized expertise, faster service, 24/7 support
4. **Pricing:** Hourly rates, fixed price, package deals, value-based
5. **Promotion:** Before/after photos, testimonials, certifications, free consultations

3. Promotional Strategies

A) EDUCATIONAL MARKETING

- **Workshops/Seminars:** Free community sessions
- **Content:** Blog posts, social media tips, videos, guides
- **Site Visits:** Show completed installations

B) COMPARATIVE MARKETING

10-Year Comparison:

- **Generator Total:** ₦4,550,000
- **Solar Total:** ₦1,200,000
- **SAVINGS:** ₦3,350,000

C) SOCIAL PROOF MARKETING

- Customer testimonials
- Case studies with photos/data
- References and referral programs
- Media coverage

D) DIGITAL MARKETING

Channels:

- Social Media (Facebook, Instagram, WhatsApp Business)
- Website with services, prices, gallery
- Google My Business
- Email marketing
- Online ads (Facebook, Google)

E) PARTNERSHIP MARKETING

Partner with electricians, contractors, real estate agents, suppliers, community organizations

F) PROMOTIONAL CAMPAIGNS

- Seasonal: "Beat the Heat," "Power Your Christmas"
- Event-based: Fuel price response, special occasions
- Customer acquisition: Referral rewards, first-customer discounts
- Retention: Free maintenance, upgrade discounts

4. Advertising Methods

A) TRADITIONAL ADVERTISING

Print: Newspapers (₦20,000-₦100,000), Flyers (₦20,000/1000)

Outdoor:

- Billboards (₦100,000-₦500,000/month)
- Banners (₦5,000-₦20,000)
- Vehicle Branding (₦30,000-₦100,000 one-time)
- Signboards (₦15,000-₦80,000)

Broadcast: Radio (₦5,000-₦30,000/spot), TV (₦50,000-₦500,000+)

B) DIGITAL ADVERTISING

Social Media Ads: Start ₦5,000, excellent targeting

Google Ads: ₦50-₦500/click, reaches ready buyers

WhatsApp Business: Free, excellent for customer service

C) GRASSROOTS MARKETING

- Word of mouth (Free, most effective)
- Community engagement
- Demonstrations and open houses

EFFECTIVENESS RANKING:

1. **Word of Mouth** - Free, highest impact
2. **Vehicle Branding** - One-time cost, continuous visibility
3. **Social Media** - Low cost, good reach
4. **Signboard** - Low cost, local visibility
5. **Flyers** - Low cost, direct targeting
6. **Radio** - Moderate cost, broad awareness
7. **Demonstrations** - Low cost, builds trust

Measuring Effectiveness

Track: Inquiries, inquiry source, conversion rate, cost per customer, ROI

ROI Formula: $(\text{Revenue} - \text{Cost}) \div \text{Cost} \times 100$

Example: ₦10,000 ad → ₦500,000 revenue = 4,900% ROI

EVALUATION

1. Define marketing and how it differs from selling.
2. Compare marketing solar goods with services.
3. List SIX strategies for promoting solar products.
4. Why is educational marketing important?
5. Compare traditional vs digital advertising.
6. Explain word of mouth marketing and its effectiveness.
7. Design a social media post for solar product/service.
8. How do you measure advertising effectiveness?

ASSIGNMENT

1. **Create marketing plan** (2-3 pages): Target customers, objectives, 5+ strategies, advertising methods, budget, success metrics.
2. **Design three ads:** Flyer, social media post, 30-second radio script.
3. **Research three solar businesses.** Compare marketing approaches, identify most effective and why.

WEEK 9: SAFETY IN SOLAR INSTALLATION AND MAINTENANCE

1. Workplace Safety Practices

Safety means maintaining a hazard-free environment to prevent accidents, injuries, or fatalities.

Importance: Solar work involves electrical hazards, heights, heavy materials, weather exposure, and costly risks.

Key Principles:

- **Planning:** Assess site, identify hazards, plan precautions, check tools/PPE, brief team.
- **PPE:** Helmet, goggles, gloves, boots, harness, vest, sun protection.
- **Site Organization:** Clear work area, organize tools, secure perimeter, display warning signs.
- **Electrical Safety:** De-energize systems, use insulated tools, test circuits, cover panels, never work alone.
- **Height Safety:** Use proper ladders/harnesses, check roof strength, avoid bad weather.
- **Lifting:** Use correct posture, team lifts, secure loads.
- **Tool Safety:** Inspect before use, use correctly, avoid wet areas, store properly.
- **Heat/Sun:** Stay hydrated, rest in shade, use sunscreen, avoid midday heat.
- **Chemical Safety:** Wear acid-resistant PPE, ensure ventilation, have neutralizing agents.
- **Fire Safety:** Keep extinguisher, avoid smoking near batteries, ensure ventilation, know emergency numbers.

2. Identifying Risks and Hazards

Hazard: Anything that can cause harm.

Risk: Likelihood the hazard causes harm.

Main Hazards:

- **Electrical:** Shock, arc flash, DC voltage. Prevent with insulation, testing, PPE.
- **Height:** Falls, falling objects. Prevent with harnesses, secured ladders, helmets.
- **Mechanical:** Cuts, crush injuries. Use gloves, lift safely, wear boots.
- **Environmental:** Heat stress, bad weather. Schedule early work, rest often.
- **Chemical:** Battery acid, hydrogen gas. Use ventilation and proper PPE.
- **Ergonomic:** Strain, awkward postures. Take breaks, use proper tools.

Risk Assessment Steps:

1. Identify hazards
2. Identify who's at risk
3. Evaluate and control risks
4. Record actions
5. Review regularly

Example: Rooftop work → risk of fall → use harness, ladder safety, 2-person team → residual risk low.

3. Safety Tools and Protective Equipment

PPE: Helmet, goggles/face shield, gloves, boots, workwear, harness, ear plugs, dust mask.

Tools/Equipment: Multimeter, insulation tester, ladders/scaffolds, lifting straps, safety signs, first aid kit, fire extinguisher, lock-out/tag-out devices.

Maintenance: Inspect, clean, replace when damaged, keep records.

4. Emergency Response Procedures

Principles (P-A-S-S): Plan, Assess, Safety first, Seek help.

Emergency Numbers (Nigeria): 112 (general), 199 (fire/ambulance).

Common Emergencies:

- **Electric Shock:** Turn off power, separate victim safely, call 112, perform CPR if trained.
- **Fall from Height:** Don't move victim, call for help, control bleeding, monitor till help arrives.
- **Chemical Exposure:** Flush skin/eyes with water, remove contaminated clothes, seek medical help.
- **Fire:** Raise alarm, use CO₂ extinguisher (Pull, Aim, Squeeze, Sweep), evacuate if large.
- **Heat Stress:** Move to shade, hydrate, cool body, seek medical aid.
- **Equipment Failure:** Stop work, isolate area, report immediately.

After Emergencies:

Report, investigate, implement corrective measures, and review safety procedures.

Incident Report Includes:

Date, location, persons involved, incident summary, injuries, cause, corrective actions, lessons learned.

Evaluation

1. List 10 workplace safety practices.
2. Explain importance of planning.
3. Identify 5 common hazards.
4. Difference between hazard and risk.
5. PPE for battery work.
6. Steps for electric shock rescue.
7. What to do if acid enters eyes.
8. Why not move fall victims.
9. P-A-S-S method.
10. Contents of an incident report.

Assignment

1. Create a **Safety Checklist** (preparation, PPE, hazards, emergency contacts, post-work).
2. Design a **Safety Poster** (PPE, hazards, emergency numbers, first aid).
3. Write an **Incident Report** on a technician's fall from height.

WEEK 10: MAINTENANCE OF SOLAR TOOLS AND EQUIPMENT

1. Meaning of Maintenance

Maintenance is the process of **servicing, preserving, and caring for tools and systems** to keep them in good working condition and extend their lifespan.

Importance:

- **Tools:** Ensures efficiency, accuracy, safety, and cost savings.
- **Solar Systems:** Sustains performance, prevents breakdowns, and protects investment.
- **Poor Maintenance Leads To:** High repair costs, downtime, inefficiency, and accidents.

2. Types of Maintenance

A. Preventive Maintenance

Performed **regularly before failure** to prevent problems.

Characteristics: Planned, scheduled, and cost-effective.

Examples:

- *Tools:* Clean, oil, sharpen, calibrate.
- *Solar Systems:* Clean panels, check batteries, tighten connections.

Schedule Example:

Frequency	Activity
Daily	Clean & inspect tools
Weekly	Oil moving parts, check batteries
Monthly	Deep clean, calibrate equipment
Quarterly	Service tools, replace worn items
Annually	Full audit & certification

Benefits: Extends lifespan, saves costs, ensures safety.

B. Corrective Maintenance

Performed **after a fault or failure** to restore function.

Characteristics: Unplanned, reactive, often costly.

Examples:

- *Tools*: Replace broken parts, repair motors.
- *Solar Systems*: Replace inverter, fix cables or batteries.

Types:

- *Emergency*: Immediate repair for safety/system failure.
- *Deferred*: Can wait—minor issues.

Aspect	Preventive	Corrective
Timing	Before failure	After failure
Cost	Lower	Higher
Planning	Planned	Unplanned
Effect	Avoids downtime	Restores function

Best Practice: Combine both—strong preventive care, ready corrective action.

3. Care and Storage of Tools

A. Hand Tools:

- Clean, inspect, and lubricate regularly.
- Store in toolboxes or boards; keep dry and organized.

B. Power Tools:

- Check cords, vents, and blades; lubricate moving parts.
- Use original cases, charge batteries properly, store in dry cabinets.

C. Electrical Testing Tools:

- Calibrate multimeters annually.
- Keep probes and screens clean; store in protective cases.

D. Ladders & Climbing Equipment:

- Inspect rungs, rails, and feet before use.
- Store dry and secure; never improvise repairs.

E. Safety Equipment:

- Inspect harnesses, hard hats, and goggles before use.
- Clean properly and replace after damage or expiry.

F. Storage Best Practices:

Organized, secure, dry, and ventilated environment with tool inventory and sign-out systems.

4. Replacing Worn-Out or Damaged Tools

When to Replace:

- Safety hazards (cracked handles, faulty insulation).
- Poor performance or excessive wear.
- Repair cost exceeds 50% of replacement.

Process: Assess → Compare costs → Decide → Document.

Guidelines:

- Replace safety gear immediately.
- Repair power tools if minor; replace if major.
- Budget 5–10% of tool value yearly for replacement.

Expected Lifespan:

Tool	Lifespan
Hand Tools	5–10 yrs
Power Tools	3–7 yrs
Multimeter	5–10 yrs
Ladders	10–15 yrs
Safety Harness	5 yrs or after fall
Hard Hats	2–5 yrs

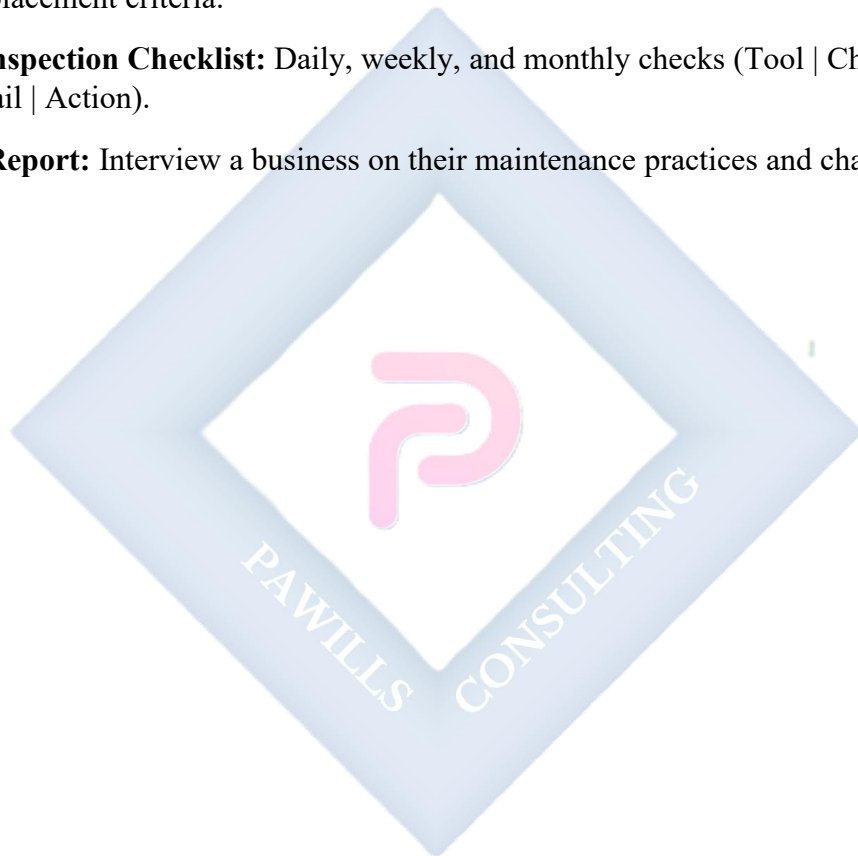
Evaluation

1. Define maintenance and its importance.
2. Compare preventive vs corrective maintenance.
3. List 8 maintenance tasks for power tools.
4. Describe proper storage for hand tools.
5. Explain care of a multimeter.
6. Safety checks before using a ladder.

7. When should a tool be replaced?
8. Design a preventive maintenance schedule.
9. Why is tool accountability important?
10. What factors guide repair vs replacement?

Assignment

1. **Tool Maintenance Manual:** Include 15 tools, maintenance steps, frequency, problems, and replacement criteria.
2. **Tool Inspection Checklist:** Daily, weekly, and monthly checks (Tool | Check Item | Pass/Fail | Action).
3. **Field Report:** Interview a business on their maintenance practices and challenges.



WEEK 11: REVISION

Comprehensive Review of Second Term Topics

Topics Covered This Term:

1. Review of First Term (Week 1)

- Solar components recap
- Load sizing review
- Safety precautions
- Common handling errors

2. Entrepreneurship (Week 2)

- Meaning and importance
- Entrepreneur characteristics
- Opportunities in solar energy

3. Concept of Money (Week 3)

- Functions of money
- Importance in business
- Capital sources
- Money management

4. Bookkeeping (Week 4)

- Meaning and importance
- Basic business records
- Recording income/expenditure
- Daily sales records

5. Business Start-Up (Week 6)

- Steps to start solar business
- Identifying customer needs
- Marketing basics
- Business ethics and customer relations

6. Marketing (Week 8)

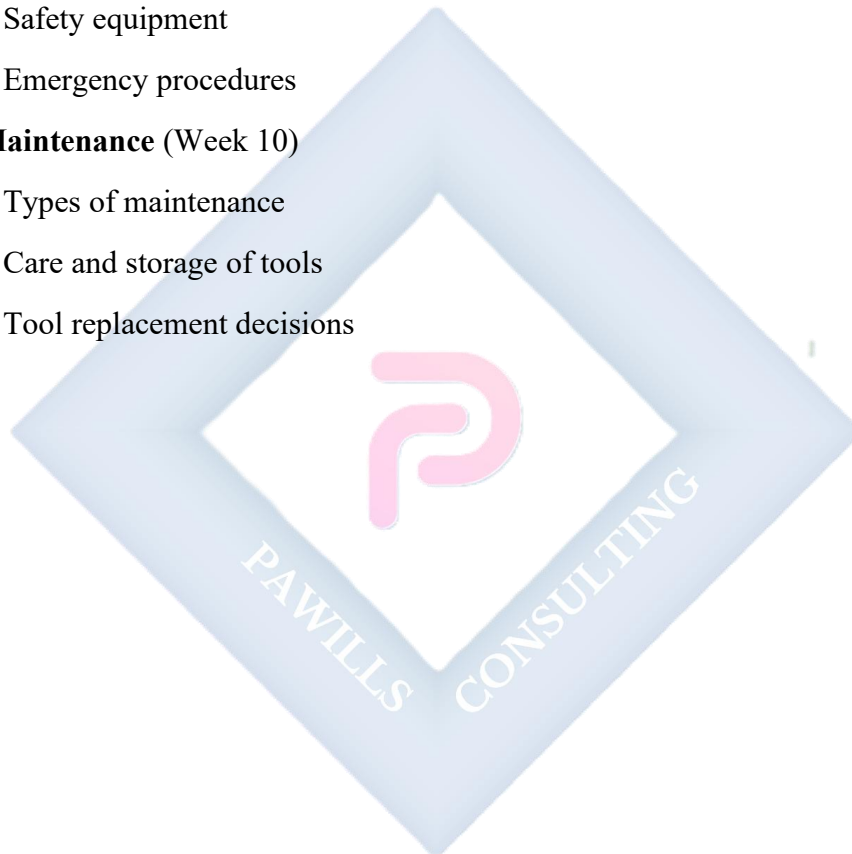
- Meaning of marketing
- Marketing goods vs. services
- Promotion strategies
- Advertising methods

7. Safety in Installation (Week 9)

- Workplace safety practices
- Risk and hazard identification
- Safety equipment
- Emergency procedures

8. Tool Maintenance (Week 10)

- Types of maintenance
- Care and storage of tools
- Tool replacement decisions



**THIRD TERM - JSS 1 SOLAR PHOTOVOLTAIC (PV) INSTALLATION AND
MAINTENANCE LESSON NOTES**



WEEK 1: REVIEW OF SECOND TERM WORK

1. Recap of Entrepreneurship and Bookkeeping

A) ENTREPRENEURSHIP REVIEW

Definition: Starting and running a business by identifying opportunities, taking risks, and organizing resources for profit.

Importance in Solar Energy:

- Creates employment
- Provides income/wealth
- Makes clean energy accessible
- Economic growth
- Environmental sustainability
- Develops local capacity

Characteristics of Good Entrepreneurs: Self-confidence, hard work, risk-taking, creativity, honesty, planning ability, communication, financial management, problem-solving, customer focus

Opportunities: Installation services, product sales, maintenance, training/consulting, manufacturing, monitoring, specialized applications

B) BOOKKEEPING REVIEW

Definition: Systematic recording and maintaining of financial transactions.

Importance: Tracks money, provides financial info, helps decisions, required for taxes, shows profitability

Basic Books:

1. **Cash Book:** Cash in/out, balance
2. **Sales Book:** Customer, product/service, amount, receipt no.
3. **Purchase Book:** Supplier, items, amount, invoice no.
4. **Debtors Book:** Money owed to business
5. **Creditors Book:** Money business owes
6. **Stock Book:** Inventory tracking

Key Principles: Record immediately, keep documents, check accuracy, separate business/personal, maintain consistency, back up records

2. Review of Marketing Strategies

Definition: Identifying, anticipating, and satisfying customer needs profitably.

The 4 Ps:

1. **Product:** What you sell (quality, features)
2. **Price:** How much you charge (competitive, profitable)
3. **Place:** Where customers find you (location, online)
4. **Promotion:** How you communicate (advertising, social media)

Marketing Strategies:

A) Educational Marketing: Workshops, content creation, demonstrations

B) Comparative Marketing: Show solar vs generator costs, environmental benefits

C) Social Proof: Testimonials, case studies, references

D) Digital Marketing: Social media, website, Google My Business, email, online ads

E) Partnership Marketing: Collaborate with electricians, contractors, agents

F) Promotional Campaigns: Seasonal offers, event-based promotions, loyalty rewards

Most Effective Methods (Ranked):

1. Word of mouth (free, highest impact)
2. Vehicle branding (continuous visibility)
3. Social media (good reach, affordable)
4. Business signboard (local visibility)
5. Flyers/door-to-door (direct, targeted)
6. Local radio (broad awareness)
7. Demonstrations (builds trust)

Measuring Effectiveness: Track inquiries, sources, conversion rate, cost per customer, ROI = $(\text{Revenue} - \text{Cost}) \div \text{Cost} \times 100$

3. Assessment of Safety and Maintenance

A) SAFETY PRACTICES

Planning & Preparation: Site assessment, risk analysis, equipment check, weather check, team briefing

PPE Required: Hard hat, safety goggles, gloves, safety boots, harness (above 2m), vest, sun protection

Safety Areas:

1. **Electrical:** De-energize, insulated tools, test first, cover panels, respect DC, never work alone
2. **Height:** Proper ladders, harness above 2m, check roof, no bad weather work
3. **Battery:** Acid-resistant PPE, ventilation, no smoking, correct connections, neutralizing solution

Common Hazards & Prevention:

- Electric shock → De-energize, PPE
- Falls → Harness, proper equipment
- Cuts → Gloves, careful handling
- Heat stress → Hydration, breaks
- Acid exposure → PPE, first aid
- Fire → Extinguisher, no smoking

Emergency Procedures:

Electric Shock:

1. Switch off power
2. Use non-conductive object if needed
3. Call 112
4. Check breathing/pulse
5. CPR if trained
6. Treat for shock

Fall:

1. Call 112
2. Do NOT move victim
3. Check consciousness/breathing

4. Control bleeding
5. Keep warm
6. Monitor until help arrives

B) MAINTENANCE PRACTICES

1. Preventive Maintenance:

- Scheduled, regular
- Prevents breakdowns
- Proactive, less expensive
- Examples: Clean panels (2-3 weeks), check battery water (monthly), tighten connections (quarterly)

2. Corrective Maintenance:

- After failure
- Repairs/replaces failed parts
- Unplanned, more expensive
- Examples: Replace failed inverter, repair damaged cables, fix loose connections

Tool Care:

Hand Tools: Clean after use, inspect regularly, lubricate, sharpen, organize storage

Power Tools: Check cords, clean vents, lubricate, replace brushes, protective cases

Testing Equipment: Clean probes, protect displays, replace batteries, annual calibration

Ladders: Inspect before use, clean, repair/replace if damaged, store horizontally

Replace When: Safety concern, damaged beyond safe use, repair cost >50% replacement, limited remaining life

EVALUATION

1. Define entrepreneurship and list FIVE characteristics of a good entrepreneur.
2. What is bookkeeping and why is it important?
3. Name and explain FOUR basic business books.
4. Calculate profit: Income ₦950,000, Expenses ₦650,000.

5. List and explain the 4 Ps of marketing.
6. What are THREE most effective marketing methods for small solar business?
7. State TEN workplace safety practices.
8. Compare preventive and corrective maintenance.
9. Describe emergency procedure for electric shock.
10. When should a tool be replaced vs repaired?

ASSIGNMENT

1. **Bookkeeping Scenario:** Design 15 transactions for one week. Record in appropriate books (cash, sales, purchases, debtors, creditors). Calculate weekly profit. Present neatly.
2. **Marketing Plan** (2 pages): Target customers (3 types), objectives, strategies (6+), budget allocation, success metrics.
3. **Safety Manual** (3-4 pages): Pre-work checklist, PPE for tasks, emergency procedures (3 scenarios), tool maintenance schedule. Professional format.



WEEK 2: INTRODUCTION TO RENEWABLE ENERGY

1. Meaning of Renewable Energy

Definition: Energy from natural sources that are continuously replenished and do not deplete when used.

Key Characteristics:

- **Naturally replenished:** Nature constantly replaces what's used
- **Sustainable:** Can be used indefinitely
- **Clean:** Produces little/no pollution
- **Abundant:** Available globally in large quantities
- **Free fuel:** After setup, energy source costs nothing

Simple Analogy: Like a fruit tree - pick fruits (use energy), tree keeps producing more. Tree doesn't disappear, unlike buried treasure (fossil fuels) that runs out.

2. Renewable vs Non-Renewable Energy

NON-RENEWABLE ENERGY

Definition: Energy from sources in limited quantities that cannot be naturally replenished within human timescale.

Examples: Coal, petroleum (petrol/diesel), natural gas, uranium

Characteristics:

- Finite (limited quantities)
- Formed over millions of years
- Depleting as we use them
- Polluting (harmful emissions)
- Non-sustainable
- Concentrated in specific locations

Problems:

- **Environmental:** Air pollution, climate change, acid rain, oil spills, mining damage
- **Economic:** Price volatility, dependency, costly extraction, finite supply
- **Health:** Respiratory diseases, cancer, asthma, cardiovascular problems

RENEWABLE ENERGY

Definition: Energy from sources naturally and continuously replenished.

Examples: Solar, wind, hydroelectric, biomass, geothermal, tidal/wave

Characteristics:

- Infinite (never runs out)
- Clean (minimal emissions)
- Sustainable indefinitely
- Distributed (available everywhere)
- Safe (no catastrophic accidents)
- Job creation

COMPARISON TABLE:

Aspect	Non-Renewable	Renewable
Supply	Limited, depleting	Unlimited, continuous
Formation	Millions of years	Constantly available
Environmental Impact	High pollution, climate change	Clean, minimal impact
Cost Trend	Increasing	Decreasing
Availability	Concentrated regions	Available everywhere
Fuel Cost	Ongoing (buy fuel)	None (free)
Sustainability	Cannot sustain future	Sustainable indefinitely
Health Impact	Causes diseases	Minimal
Examples	Coal, oil, gas	Solar, wind, water

3. Advantages of Renewable Energy

A) ENVIRONMENTAL BENEFITS

1. **Reduces Air Pollution:** No smoke/toxic emissions, cleaner air, healthier communities
2. **Fights Climate Change:** No greenhouse gases, slows global warming

3. **Protects Water Resources:** Doesn't pollute water, minimal usage, no spills
4. **Preserves Habitats:** No mining/drilling, protects wildlife, maintains biodiversity

B) ECONOMIC BENEFITS

1. **Reduces Energy Costs:** Free fuel after installation, no monthly bills
 - Example: Generator ₦30,000/month = ₦3,600,000/10 years vs Solar ₦0 fuel
2. **Creates Jobs:** Manufacturing, installation, maintenance, sales, R&D
3. **Energy Independence:** Reduces reliance on imports, strengthens security
4. **Stable Prices:** Free sun/wind, predictable costs
5. **Rural Development:** Brings electricity to remote areas economically

C) SOCIAL BENEFITS

1. **Reliable Electricity:** Consistent power, not dependent on fuel/PHCN
2. **Health Improvements:** Cleaner air, powered healthcare, medicine refrigeration
3. **Educational Advancement:** Schools powered, students study at night, access to technology
4. **Economic Opportunities:** Reliable business operations, home-based businesses
5. **Gender Equality:** Less time collecting firewood, more education opportunities

D) TECHNICAL BENEFITS

1. **Scalability:** Small to large systems, add as needed, modular
2. **Low Maintenance:** Few moving parts, 20-25+ years lifespan
3. **Silent Operation:** No noise pollution
4. **Quick Installation:** Operational in days/weeks
5. **Distributed Generation:** Power where used, reduces transmission losses

E) STRATEGIC BENEFITS

1. **Energy Security:** Not vulnerable to supply disruptions
2. **Risk Reduction:** No catastrophic failures, safer overall
3. **Technology Advancement:** Drives innovation, reduces costs
4. **International Relations:** Reduces conflicts, promotes cooperation

4. Sources of Renewable Energy

A) SOLAR ENERGY

Source: Sun's radiation

How It Works: Sun emits radiation → Solar panels capture → Photovoltaic cells convert to electricity

Advantages: Abundant worldwide, available daily, no moving parts, silent, scalable, 20-25+ years lifespan

Applications: Home systems, street lights, water pumping, phone charging, calculators, satellites

Nigeria's Potential: Near equator, 5-7 peak sun hours daily, year-round availability, huge untapped potential

B) WIND ENERGY

Source: Moving air (kinetic energy)

How It Works: Wind turns turbine blades → Connected to generator → Produces electricity

Advantages: Clean, cost-effective in windy areas, land beneath usable, quick installation, scalable

Applications: Wind farms, small home turbines, offshore farms, water pumping

Nigeria's Potential: Northern regions, coastal areas, Jos Plateau, growing interest

Disadvantages: Requires consistent wind, noise, affects birds, visual impact, variable output

C) HYDROELECTRIC ENERGY

Source: Flowing/falling water

How It Works: Water flows through turbines → Pressure turns blades → Drives generators → Produces electricity

Types: Large hydro (Kainji, Shiroro), small hydro, micro hydro, run-of-river

Advantages: Reliable, 50-100 years lifespan, stores water for peak demand, flood control, irrigation

Applications: Large-scale generation, village supplies, irrigation pumping

Nigeria's Potential: Major rivers (Niger, Benue), existing dams, untapped small-hydro potential

Disadvantages: Can displace communities, affects ecosystems, high initial cost, dependent on water flow

D) BIOMASS ENERGY

Source: Organic materials (plants, animals, waste)

Types:

- Traditional: Firewood, charcoal, animal dung
- Modern: Biogas, biofuels (ethanol, biodiesel), pellets

How It Works: Direct combustion, anaerobic digestion (biogas), fermentation (ethanol), gasification

Advantages: Uses waste, reduces landfill, carbon-neutral, readily available, supports agriculture

Applications: Cooking fuel (biogas), electricity, vehicle fuel, heating

Nigeria's Potential: Large agriculture (crop residues), livestock (animal waste), municipal waste, forests

Disadvantages: Can cause deforestation, some emissions, land use conflicts, traditional use inefficient

E) GEOTHERMAL ENERGY

Source: Earth's interior heat

How It Works: Earth's core heat → Heats underground water → Steam/hot water to surface → Drives turbines

Advantages: Constant 24/7, small footprint, minimal emissions, 30+ years lifespan

Applications: Electricity, district heating, greenhouse heating, industrial processes

Nigeria's Potential: Limited, some hot springs, not a major source

F) TIDAL AND WAVE ENERGY

Source: Ocean tides and waves

How It Works: Tides/waves turn turbines → Generate electricity

Advantages: Predictable, high energy density, no fuel, long life

Applications: Coastal electricity, island power

Nigeria's Potential: 853 km coastline, some tidal variation, technology developing, not widely used

Disadvantages: High initial cost, limited to coastal areas, affects marine ecosystems, saltwater corrosion

RENEWABLE ENERGY RANKING FOR NIGERIA:

1. **Solar** - Excellent (abundant sunshine nationwide)
2. **Biomass** - Very good (agricultural waste, biogas)
3. **Hydro** - Good (existing infrastructure, expansion possible)
4. **Wind** - Moderate (northern/coastal regions)
5. **Geothermal** - Limited
6. **Tidal/Wave** - Limited

Recommendation: Nigeria should prioritize solar and biomass due to abundant resources and suitability for distributed generation.

EVALUATION

1. Define renewable energy and state FOUR characteristics.
2. Explain difference between renewable and non-renewable with THREE examples each.
3. Why are fossil fuels non-renewable?
4. List and explain SIX advantages of renewable energy.
5. Describe how solar energy converts to electricity.

ASSIGNMENT

1. How does wind energy work?
2. What is biomass energy? Give FOUR examples.
3. Compare hydroelectric and solar energy (THREE similarities, THREE differences).
4. Why is solar energy suitable for Nigeria?
5. What are TWO disadvantages each of: wind, hydroelectric, and biomass energy?

WEEK 3: PRINCIPLES OF SOLAR ENERGY

1. Meaning of Solar Energy

Solar energy is the light and heat from the sun converted into useful energy, such as electricity and heat.

Facts about the Sun:

- Distance from Earth: 150 million km
- Age: 4.6 billion years
- Surface temperature: 5,500°C
- Core temperature: 15 million°C
- Main elements: Hydrogen (74%) and Helium (24%)

Energy Production:

The sun produces energy through **nuclear fusion** — hydrogen atoms combine to form helium, releasing enormous energy that travels to Earth as sunlight (takes about 8 minutes).

Energy Reaching Earth:

The sun provides more energy in one hour than humans use in a year, though only a small fraction reaches Earth.

Solar Spectrum:

Sunlight contains visible light, infrared (heat), and ultraviolet (causes sunburn). Solar panels mainly use visible and infrared light.

2. How Solar Energy is Generated

The Photovoltaic Effect:

This is the process where light energy produces electric current in certain materials.

Steps:

1. Solar Cell Construction:

- Made of silicon layers — **N-type** (extra electrons) and **P-type** (electron holes) forming a **P-N junction**.

2. Sunlight Strikes Panel:

- Photons excite electrons, freeing them to move.

3. Electric Field Forms:

- The P-N junction directs electrons, creating voltage.

4. Current Flows:

- Metal contacts collect electrons; current (DC electricity) flows through an external circuit.

Factors Affecting Solar Power:

1. **Sunlight intensity** – strongest at midday.
2. **Temperature** – cooler panels work better.
3. **Angle & Orientation** – face south, tilt about Nigeria's latitude.
4. **Cleanliness** – dust or dirt reduces efficiency.
5. **Panel quality** – better panels are more efficient.

3. Conversion of Sunlight to Electricity

Stage	Component	Function
1	Solar Panels	Convert sunlight to DC power
2	Charge Controller	Regulates voltage to batteries
3	Battery	Stores energy for later use
4	Inverter	Converts DC to AC for home use
5	Distribution	Supplies electricity to appliances

Efficiency:

Only about **10–15%** of sunlight becomes usable AC power due to system losses, but it's clean and renewable.

4. Importance of Solar Power in Daily Life

A. Home Use:

- Lighting, fans, and appliances
- Food preservation (solar fridges)
- Device charging and communication
- Water pumping and cooking

B. Education:

- Electrified classrooms and computer labs
- Improved student and teacher productivity

C. Health:

- Power for clinics, vaccine storage, and night operations

D. Economy:

- Reduces fuel costs
- Powers small businesses and farms
- Creates solar installation jobs

E. Environment:

- Reduces air pollution and carbon emissions
- Protects forests (less firewood use)

F. Energy Independence:

- No fuel or PHCN worries
- Predictable and long-term power supply

Example:

Ade's family in Lagos switched to solar:

- Saved ₦30,000/month on fuel
- Reliable, quiet power 24/7
- Paid off system in just over 2 years

Evaluation

1. Define solar energy.
2. Explain the photovoltaic effect.
3. Mention and describe two silicon layers in a solar cell.
4. List five factors affecting solar energy generation.
5. Describe how sunlight becomes electricity.
6. State two types of charge controllers and their differences.
7. Why is solar conversion not 100% efficient?
8. Calculate energy from a 350W panel in Kano (6 hours).
9. List five benefits of solar energy in daily life.
10. How does solar energy protect the environment?

Assignment

1. Draw and label the structure of a solar cell.
2. Write a short case study of a home or business that switched to solar.
3. Design a simple poster titled *“From Sunlight to Electricity.”*



WEEK 4: SOLAR PV SYSTEM COMPONENTS

1. Main Components Review

A) Solar Panels

Function: Convert sunlight to DC electricity

Key Specifications:

- **Power Rating:** 100W-400W+ (e.g., 300W at peak sun)
- **Voltages:**
 - V_{oc} (Open Circuit): 36-45V
 - V_{mp} (Max Power): 30-36V
- **Currents:**
 - I_{sc} (Short Circuit): 8-10A
 - I_{mp} (Max Power): 8-9A
- **Efficiency:** Mono 19-22%, Poly 15-18%, Thin Film 10-13%
- **Physical:** $\sim 1.6m \times 1m$, 15-25kg, aluminum frame

Installation: Angle = latitude ($6-13^\circ$ Nigeria), face south, series for voltage, parallel for current

B) Batteries

Function: Store energy for later use

Key Specifications:

- **Voltage:** 2V, 6V, 12V cells (series for higher voltage)
- **Capacity:** Amp-hours (Ah) - e.g., $200Ah \times 12V = 2.4kWh$
- **DoD (Depth of Discharge):** Lead-acid 50%, Lithium 80-90%
- **Cycle Life:** Lead-acid 500-1,500, Lithium 3,000-5,000

Types:

- **Flooded:** Cheapest, needs maintenance, can spill
- **AGM:** Sealed, no maintenance, costlier
- **Gel:** Best deep discharge, slow charging
- **Lithium:** Lightest, longest life, most expensive

Connections:

- **Series:** Voltages add ($12V+12V=24V$), same capacity
- **Parallel:** Capacity adds ($200Ah+200Ah=400Ah$), same voltage

C) Inverters

Function: Convert DC to AC electricity

Key Specifications:

- **Power:** Continuous (2000W) + Surge ($2\times$ for motors)
- **Input:** Must match battery (12V, 24V, 48V)
- **Output:** 220-240V, 50Hz (Nigeria)
- **Efficiency:** 90-95%

Types:

- **Pure Sine Wave:** Works with all appliances, recommended
- **Modified Sine Wave:** Cheaper, limited compatibility

Protection: Overload, short circuit, low/high voltage, over-temperature

D) Charge Controllers

Function: Regulate charging, prevent overcharge

Key Specifications:

- **Current Rating:** Match panel output + 25% safety
- **Voltage:** 12V, 24V, or 48V systems

Types:

Feature	PWM	MPPT
Cost	₦8,000-25,000	₦25,000-100,000+
Efficiency	70-80%	92-98%
Best for	Systems <500W	Systems >500W
Panel voltage	Must match battery	Can be much higher

Charging Stages: Bulk → Absorption → Float → Equalization

2. System Connection

Wiring Flow:

PANELS → CHARGE CONTROLLER → BATTERY → INVERTER → LOADS

Connection Sequence (CRITICAL ORDER):

Step 1: Battery Bank

- Position on stable, non-conductive surface
- Connect for desired voltage/capacity
- Use thick cables ($\geq 16\text{mm}^2$)
- Install fuses on positive cables

Step 2: Charge Controller

1. **Battery first:** Negative, then positive (through fuse)
2. **Loads:** If controller has load terminals
3. **Panels last:** Cover panels first, connect, then uncover

Step 3: Solar Panels

- Series: Positive to negative (voltages add)
- Parallel: All positives together (current adds)
- Use combiner box with fuses per string
- UV-resistant cables in conduit

Step 4: Inverter

- Very thick cables (50mm^2+) to battery
- Fuse near battery (200-400A)
- AC output to distribution board
- Keep close to batteries

Step 5: Grounding

- All metal frames to ground rod
- One DC point grounded
- AC system properly earthed

Step 6: Protection

- Fuses: panels, battery-controller, battery-inverter
- Breakers: AC circuits

- Surge protection for lightning

3. Fault Diagnosis

Common Faults:

No Power from Panels

- Check: Shading, connections, fuses, cables
- Test: Voc at panel (36-45V expected)

Battery Not Charging

- Check: Battery voltage, connections, fuses, controller settings
- Test: Voltage at battery vs. controller terminals

Inverter Won't Start

- Check: Battery voltage, overload, DC connections
- Test: Error codes, voltage at inverter input

Rapid Battery Drain

- Check: Load size, battery age, parasitic drains
- Test: Capacity test, current draw with loads off

Low Solar Production

- Check: Dirt, shading, panel angle, temperature
- Test: Voc and Isc vs. datasheet values

Systematic Approach:

1. Gather symptoms
2. Visual inspection
3. Check simple things (fuses, connections)
4. Isolate problem area
5. Test systematically from source to load
6. Measure voltages at each point
7. Document and verify fix

4. Mini-Solar Setup (50W System)

Components:

- 50W solar panel
- 12V 40Ah battery
- 10A PWM controller
- 3× 10W LED bulbs
- Cables, fuses, switch

Assembly Steps:

1. **Inspect** all components
2. **Connect battery** to controller (negative first)
3. **Mount panel** facing south, 10° angle
4. **Connect panel** (covered, then uncover)
5. **Wire LED bulbs** in parallel
6. **Test** charging and load operation

Performance:

- Load: 30W = 2.5A at 12V
- Runtime: $40\text{Ah} \div 2.5\text{A} = 16$ hours theoretical
- Safe runtime (50% DoD): **8 hours**
- Cost: ~~~N~~59,000
- Payback: ~18 months

Key Safety Rules:

- Battery before panels
- Panel covered during connection
- Proper fusing throughout
- Adequate ventilation
- One-hand rule when testing live circuits

EVALUATION

1. List the four main components of a solar PV system and state the function of each.
2. What are the three voltage ratings typically indicated on a solar panel label?
3. Explain the difference between series and parallel connection of batteries. When would you use each?
4. Compare PWM and MPPT charge controllers (give THREE differences).
5. Why must batteries be connected to the charge controller BEFORE connecting solar panels?

ASSIGNMENT

1. Document your participation in the mini-solar setup assembly:

- Draw detailed wiring diagram of the complete system
- List all components used with specifications
- Describe step-by-step what you did during assembly
- Record initial test measurements
- Explain any problems encountered and solutions
- Include photos or sketches

2. For each scenario, explain the diagnosis process and solution:

Scenario A: System was working fine. After cleaning the panels, system produces no power. Panels show voltage but controller shows 0A charging current.

Scenario B: Customer complains their new system only powers lights for 2 hours at night, but installer promised 6 hours. System has 100Ah battery and 100W of lights.

Scenario C: Inverter keeps shutting down with overload error whenever refrigerator compressor starts, even though inverter is rated at 1500W and refrigerator is only 120W.

For each:

- List possible causes
- Explain diagnostic steps
- Provide solution

WEEK 6: ENERGY STORAGE SYSTEMS

1. Understanding Energy Storage

Definition:

Energy storage is the process of capturing energy for later use. In solar PV systems, it stores power produced during the day for use at night or when sunlight is weak.

Why It's Needed:

- Solar energy is not constant.
- Balances energy supply and demand.
- Ensures continuous power even at night or during grid failure.

Simple Analogy: Like a water tank that stores rainwater for use when there's no rain.

Types of Energy Storage:

- **Batteries (main type in solar systems)**
- Pumped hydro, compressed air, flywheel, thermal, and hydrogen (for large-scale applications).

2. Functions of Batteries in Solar Systems

Primary Functions:

1. Stores excess energy from solar panels.
2. Provides power at night or during low sunlight.
3. Stabilizes voltage and power supply.
4. Acts as backup during outages.
5. Enables off-grid operation.

Additional Functions:

- Reduces peak demand costs.
- Maintains frequency stability.
- Integrates renewable sources smoothly.

Key Battery Terms:

- **Capacity (Ah):** Amount of charge stored.
- **Depth of Discharge (DoD):** Portion of energy that can be used safely.

- **Cycle Life:** Number of charge and discharge cycles.
- **Efficiency:** How well the battery converts stored energy.
- **Temperature:** Ideal range is 20–25°C.

3. Types of Solar Batteries

A. Lead-Acid Batteries

- **Flooded (Wet Cell):** Low cost, regular maintenance (add water).
- **AGM:** Sealed, maintenance-free, higher cost.
- **Gel:** Sealed, deep discharge, longer lifespan, expensive.

Type	Cost	Maintenance	Cycle Life	DoD
Flooded	Low	High	500–1000	50%
AGM	Medium	None	800–1200	50%
Gel	High	None	1200–1500	60–80%

B. Lithium-Ion (LiFePO₄)

- Lightweight, fast charging, and 10–15 years lifespan.
- Deep discharge (85–90%), no maintenance.
- Higher initial cost but best long-term value.

4. Battery Maintenance and Safety

For Flooded Lead-Acid Batteries:

- **Monthly:** Check and refill with distilled water.
- **Clean terminals** and check voltage regularly.
- **Avoid overcharging** or complete discharge.

For AGM/Gel/Lithium:

- Minimal maintenance; just keep terminals clean and monitor voltage or BMS.

Safety Precautions:

- Use **goggles, gloves, boots, and apron.**
- Ensure **good ventilation.**

- Avoid **smoking or sparks** near batteries.
- Handle carefully — batteries are heavy and corrosive.

In Case of Emergency:

- **Acid on skin:** Wash with water for 15 minutes.
- **Eye contact:** Rinse immediately and seek help.
- **Fire:** Use a Class D (for lithium) or ABC (for lead-acid) extinguisher.

Evaluation

1. Define energy storage and its importance in solar systems.
2. List five functions of batteries in a solar system.
3. Differentiate between Flooded, AGM, and Gel batteries.
4. Explain the meaning of Depth of Discharge (DoD).
5. State three safety precautions during battery maintenance.
6. Mention two advantages of Lithium-ion batteries.

Assignment

1. **Illustrate** a simple diagram of a solar energy storage system showing the flow of energy from panels to batteries, inverter, and loads.
2. **Write a short report (1 page)** comparing two families or businesses — one using a generator and another using solar with battery storage. Highlight differences in cost, comfort, and reliability.
3. **Create a mini chart** showing the life span, efficiency, and DoD of the four major battery types (Flooded, AGM, Gel, and Lithium-ion).

WEEK 8: POWER DISTRIBUTION AND LOAD MANAGEMENT

1. Power Distribution

Meaning:

Power distribution is the process of delivering electricity from the inverter to different appliances safely and efficiently.

System Flow:

Solar Panels → Charge Controller → Batteries → Inverter → Distribution Board → Circuits → Loads

Distribution Board:

- Central point for dividing power into circuits
- Provides protection (via breakers)
- Allows maintenance isolation

Main Components:

- **Main Breaker:** Controls total power, serves as emergency shutoff
- **Branch Breakers:** Protect individual circuits (lighting, sockets, kitchen, etc.)

Typical Circuits:

- Lighting (10–15A)
- Sockets (15–20A)
- Kitchen (20–30A)
- Air Conditioner (30A+)
- Water Pump (20–30A)

Why Separate Circuits?

For safety, convenience, performance, and compliance with electrical codes.

Cable Sizing (Nigeria):

1.5mm² (lights), 2.5mm² (sockets), 4–6mm² (heavy loads), 10mm²+ (main).

2. Understanding DC and AC Power

DC (Direct Current):

- Flows in one direction, constant voltage
- Produced by solar panels and batteries

- Easier storage, better for electronics
- Hard to transform, limited appliance use

AC (Alternating Current):

- Flows in both directions, changes polarity
- 50Hz frequency in Nigeria
- Easy voltage transformation
- Compatible with most household appliances
- Cannot be stored directly

Key Differences:

Feature	DC	AC
Flow	One direction	Alternates
Voltage	Constant	Variable
Storage	Easy	Not possible
Transformation	Difficult	Easy
Use	Electronics	Most appliances

Power Factor (PF):

Ratio of real to apparent power. Low PF causes inefficiency and requires larger inverter capacity.

3. Load Balancing

Meaning:

Distributing power evenly across circuits and time to avoid overloads and improve efficiency.

Methods:

- **Circuit Level:** Separate heavy loads, use dedicated circuits
- **Time-Based:** Schedule heavy appliances during daylight (solar peak hours)
- **Load Prioritization:**
 - *Essential:* Lights, fridge, phone chargers
 - *Important:* TV, fans
 - *Optional:* Iron, washing machine
- **Load Reduction:** Use energy-efficient devices (LEDs, inverter ACs)

- **Inverter Load Management:** Keep total running load within inverter capacity.

4. Effects of Overload and Underload

Overload:

- Too many appliances → low voltage, inverter trips, overheating, battery drain, fire risk.
Prevention: Proper sizing, load management, and protection devices.

Underload:

- Very small load on large inverter → inefficiency, wasted power, higher cost.
Solution: Add moderate loads, enable eco-mode, right-size future systems.

Optimal Load:

- Inverter runs at 40–80% capacity
- Batteries cycle between 50–80% charge
- Balanced circuits and efficient use

Evaluation

1. Define power distribution and its importance in solar systems.
2. Explain the flow of power from solar panels to appliances.
3. State two functions of a distribution board.
4. Differentiate between DC and AC current.
5. Why can't batteries store AC electricity?
6. What is power factor?
7. Mention three ways to balance electrical loads.
8. State two effects of system overload.
9. Explain what happens when a 2kW inverter powers only 100W load.
10. Calculate energy from a 12V battery supplying 15A for 6 hours.

Assignment

1. Load Management Plan:

Create a daily schedule showing when each appliance should be used (for a 3kW inverter system).

2. **Circuit Design:**

Design a distribution layout for a 3-bedroom solar home with at least 5 circuits, showing breaker ratings and cable sizes.

3. **Troubleshooting:**

Analyze why:

- (a) Lights dim when the fridge starts.
- (b) Inverter shuts down at night.
- (c) A 5kW system wastes power at low load.



WEEK 9: TROUBLESHOOTING IN SOLAR PV SYSTEMS

Meaning of Troubleshooting:

Troubleshooting is a systematic process of diagnosing, identifying, and solving faults in a solar PV system to restore normal operation. It differs from guessing, which wastes time, money, and may cause further damage.

Key Principles of Troubleshooting:

1. **Stay Calm:** Think logically; avoid panic or hasty actions.
2. **Be Systematic:** Follow clear steps; document actions; check simple things first.
3. **Use Your Senses:** Look, listen, touch (carefully), and smell for abnormalities.
4. **Ask Questions:** Find out when and how the problem started.
5. **Test Before Replacing:** Verify faults using measurements.
6. **Safety First:** Switch off power, wear PPE, and avoid risks.

Troubleshooting Process (Scientific Method):

1. Observe the problem.
2. Form a hypothesis (possible causes).
3. Test and measure.
4. Analyze results.
5. Identify root cause.
6. Fix the issue.
7. Verify the solution.

COMMON SOLAR PV SYSTEM PROBLEMS

1. No Power from System

- *Causes:* Discharged or faulty battery, inverter failure, blown fuse, or loose connections.
- *Diagnosis:* Check inverter display, measure battery/inverter voltages, inspect fuses and terminals.

2. Solar Panels Not Charging

- *Causes:* Shading, dirt, damaged panels, faulty controller, or wiring faults.
- *Diagnosis:* Inspect panels, measure open-circuit voltage (Voc), check controller input/output, verify settings.

3. Short Backup Time

- *Causes:* Weak or undersized batteries, high loads, parasitic drains.
- *Diagnosis:* Check voltage and capacity, perform load audit, test for current drains.

4. Intermittent Operation

- *Causes:* Loose connections, overheating, overloads, or faulty components.
- *Diagnosis:* Observe when issues occur, tighten connections, improve ventilation.

5. Low Solar Production

- *Causes:* Shading, dirt, poor orientation, degraded panels, wiring losses.
- *Diagnosis:* Clean panels, check angles, test panel voltage/current, inspect wiring.

6. Battery Not Holding Charge

- *Causes:* Sulfation, shorted cell, loss of electrolyte, incorrect charging.
- *Diagnosis:* Voltage and load tests, check specific gravity, verify charge settings.

7. Overheating Components

- *Causes:* Loose terminals, undersized cables, overload, poor ventilation.
- *Action:* Stop system, inspect, correct cable sizes, replace damaged parts.

8. Battery Boiling/Gassing

- *Causes:* Overcharging, high temperature, wrong controller settings.
- *Action:* Adjust voltage, improve ventilation, refill with distilled water if needed.

FAULT-FINDING PROCEDURE

1. **Information Gathering:** Ask user about system behavior and history.
2. **Visual Inspection:** Check for loose wires, corrosion, blown fuses, damaged parts.
3. **Basic Measurements:**
 - Battery voltage (12.6–12.8V = full, <11.8V = low).
 - Panel Voc ($\approx$ 36–45V per panel).
 - Controller readings (solar input/output).
 - Inverter voltage and indicators.
4. **Analyze Findings:** Compare results, form hypothesis.

5. **Targeted Testing:** Swap components, test with known good parts.
6. **Identify Root Cause:** Fix underlying fault, not just symptoms.
7. **Implement Solution:** Repair or replace faulty parts, document work.
8. **Verify Operation:** Test system under various loads; confirm stability.
9. **Follow-Up:** Recheck after a few days to ensure fault doesn't recur.

BASIC TOOLS AND REPAIRS

Essential Tools:

- **Measurement:** Multimeter, clamp meter, hydrometer, load tester.
- **Hand Tools:** Screwdrivers, wrenches, pliers, crimpers, cutters, strippers.
- **Safety:** Goggles, gloves, boots, PPE, fire extinguisher.
- **Cleaning:** Soft brush, mild soap, distilled water, cloths.
- **Documentation:** Manuals, diagrams, notebook.

Simple Repairs:

1. Replace blown fuse (after finding cause).
2. Tighten and clean corroded connections.
3. Clean dusty panels in cool weather.
4. Top up flooded batteries with distilled water.
5. Reset tripped breakers safely.

When to Call a Professional:

- Internal inverter or controller faults.
- High-voltage DC systems.
- Complex wiring or component replacements.

EVALUATION

1. Define troubleshooting and explain its importance.
2. List the major steps in fault diagnosis.
3. Mention five possible causes of a dead inverter.

4. Describe how to diagnose a “no charging” problem.
5. Explain how to safely measure battery voltage with a multimeter.

ASSIGNMENT

1. **Create a Troubleshooting Flowchart** for solar PV systems showing different fault paths and solutions.
2. **Fault Analysis Exercise:** Diagnose and explain causes for three real-world system issues (battery low, inverter overload, low solar charge).
3. **Technician Toolkit Design:** List and describe 20 essential tools, grouped into categories (measurement, safety, hand tools, etc.).



WEEK 10: FIELD PRACTICE AND DEMONSTRATION

Project Title: *Complete Installation and Testing of a Small Solar PV System*

This week provides full practical experience — from site assessment to installation, testing, and handover — applying all concepts learned throughout the term.

SYSTEM OVERVIEW

Application: Classroom or workshop power supply

System Components:

- $2 \times 100\text{ W}$ panels (or $1 \times 200\text{ W}$)
- $1 \times 12\text{ V}$, 100 Ah deep-cycle battery
- $1 \times 20\text{ A}$ PWM charge controller
- $1 \times 800\text{ W}$ pure sine inverter
- Mounting frame, cables, fuses, and protection devices

Estimated Load:

- 6 bulbs (60 W), 2 fans (120 W), 1 laptop/projector (100 W) $\rightarrow \approx 280\text{ W total}$

Expected Output:

- Solar generation $\approx 800\text{ Wh/day}$
- Consumption $\approx 1,200\text{ Wh/day}$
- Battery supports evening use

PRACTICAL ACTIVITIES (4 DAYS)

Day 1 – Planning and Preparation

- **Site Assessment:** Identify panel location, shading, orientation (south-facing), and cable routes.
- **Design Review:** Confirm load calculations, check all components, and plan installation sequence.
- **Tool Preparation:** Gather tools, cut and label cables, review safety procedures.
- **Pre-installation Testing:** Measure panel Voc ($\sim 21\text{ V}$), battery voltage (11.5–12.7 V), verify inverter and controller functionality.

Day 2 – Mechanical Installation

- **Panel Mounting:** Install roof or ground structure, fix panels securely, and route cables neatly.
- **Equipment Mounting:** Fix battery, charge controller, inverter, and distribution board in ventilated, accessible areas.
- **Safety Installations:** Grounding system, fuses, circuit breakers, and warning labels properly fitted.

Day 3 – Electrical Installation

- **DC Wiring:** Connect panels (series or parallel), route to charge controller, install fuses.
- **Battery Connection:** Install using correct polarity and cable sizes (16 mm² for inverter line).
- **System Connection Sequence:**
 1. Battery → Charge Controller
 2. Load → Controller or Inverter
 3. Solar Panels → Controller
 4. Battery → Inverter
- **AC Wiring:** Connect inverter output to distribution board; wire circuits for lights and sockets with proper earthing.

Day 4 – Testing, Commissioning, and Handover

- **Testing:**
 - Measure voltages and currents (DC: 12–13 V, AC: 220–240 V).
 - Verify charging and load operation.
 - Perform functional and protection tests (breaker trip, overload, cutoff).
- **Documentation:** Record system details, test results, cable sizes, fuse ratings, and draw schematic diagram.
- **User Training:** Explain system operation, maintenance, safety, and troubleshooting basics.
- **Final Handover:** Confirm performance, provide manuals, and obtain user acceptance.

TESTING SUMMARY

Test	Purpose	Expected Result
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Panel Voc	Verify panel health	≈ 21 V per 100 W panel
Isc	Confirm current capability	Matches datasheet
Charging Current	Check charge rate	2–4 A typical
Load Current	Verify load draw	~ 23 A at 280 W load
Inverter Efficiency	Measure performance	85–95%

MAINTENANCE SCHEDULE

- **Daily:** Visual inspection, check indicators.
- **Weekly:** Clean panels, tighten connections.
- **Monthly:** Check battery water and voltage.
- **Quarterly:** Full inspection and load testing.
- **Annually:** Professional servicing and performance review.

LEARNING OUTCOMES

By the end of this project, students should be able to:

1. Plan and install a complete solar PV system.
2. Conduct safe and correct wiring.
3. Test and commission the system.
4. Diagnose and fix faults.
5. Prepare professional documentation.
6. Train users on operation and maintenance.

EVALUATION

1. State the correct order of system connections.
2. Describe how to test a panel's open-circuit voltage.
3. Explain how to measure charging current using a clamp meter.
4. Mention three safety checks before energizing a new installation.
5. What details should be recorded in a maintenance logbook?

ASSIGNMENT

1. **Installation Report:** Document your field practice (site, procedure, test results, lessons learned).
2. **System Documentation:** Prepare a complete user package with diagram, manuals, and maintenance guide.
3. **Troubleshooting Simulation:** Create five realistic fault scenarios with symptoms, causes, and solutions.



WEEK 11: REVISION

Comprehensive Review of Third Term Topics

Topics Covered This Term:

- 1. Review of Second Term (Week 1)**
 - Entrepreneurship and bookkeeping
 - Marketing strategies
 - Safety and maintenance
- 2. Introduction to Renewable Energy (Week 2)**
 - Meaning and types
 - Renewable vs. non-renewable
 - Advantages and sources
- 3. Principles of Solar Energy (Week 3)**
 - Sun as energy source
 - Photovoltaic effect
 - Energy conversion process
 - Importance in daily life
- 4. Components of Solar PV Systems (Week 4)**
 - Detailed component specifications
 - System connections
 - Fault identification
 - Practical assembly
- 5. Energy Storage Systems (Week 6)**
 - Energy storage necessity
 - Battery functions
 - Battery types (lead-acid, lithium)
 - Maintenance and safety
- 6. Power Distribution and Load Management (Week 8)**
 - Distribution systems

- DC vs. AC power
- Load balancing strategies
- Overload and underload effects

7. **Troubleshooting** (Week 9)

- Systematic fault-finding
- Common problems
- Diagnostic procedures
- Basic repairs

8. **Field Practice** (Week 10)

- Complete system installation
- Testing procedures
- Documentation
- User training

